ACADEMIC & COMPETITIVE EXAMINATION SERIES

COMPUTER VISION

Complete Unit-Wise Question & Answer Collection, Recaps & 2-Mark Guide

SYLLABUS COVERAGE
Units 1 – 5 Complete Curriculum

QUESTIONS & SOLUTIONS
37 Long Answers + 26 Short (2M) Questions

DIAGRAMS & FORMULAS
25 Labeled Vector Diagrams + Bordered Formula Boxes

REVISION FEATURES
5 Unit Recaps + Master Topic Matrix Index

Academic Formatting Standard: This comprehensive master volume contains complete long-form academic solutions, labeled vector diagrams and flowcharts, dedicated bordered formula boxes for all mathematical equations, full unit recaps, and short-answer 2-mark question banks.

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UNIT 1

Unit 1: Fundamentals of Image Formation & Enhancement

Comprehensive Question & Answer Collection • Quick Revision Recap • 2-Mark Direct Questions

PART A: DETAILED EXAMINATION QUESTIONS & ANSWERS

Unit 1 • Q1

Define Computer Vision and List Its Key Applications.

> 📝 **Exam Question:** form now this qustions are follow the rule i have said earlier unit1 1.Define computer vision and list its key applications.

Unit 1 – Q1. Define Computer Vision and List Its Key Applications.

Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to capture, process, analyze and understand digital images and videos. It allows machines to identify objects, recognize faces, detect movements and understand their surroundings, similar to human vision.

Computer Vision uses techniques such as **Image Processing, Feature Extraction, Object Detection and Machine Learning** to extract meaningful information from visual data.

Key Applications of Computer Vision

1. **Face Recognition:** Identifies and verifies people using facial features. Example: Smartphone face unlock.
2. **Object Detection:** Detects and identifies objects in images or videos. Example: Detecting cars and pedestrians.
3. **Medical Imaging:** Helps doctors analyze X-rays, MRI and CT scans to identify diseases.
4. **Autonomous Vehicles:** Helps self-driving cars recognize roads, traffic signs, pedestrians and other vehicles.
5. **Surveillance Systems:** Monitors activities and detects suspicious events using security cameras.
6. **Industrial Automation:** Detects defective products and checks product quality in factories.
7. **Augmented Reality (AR):** Combines virtual objects with the real-world environment.
8. **Agriculture:** Identifies plant diseases and monitors crop conditions using images.

Basic Working of Computer Vision

1. **Image Acquisition:** Captures images using cameras or sensors.
2. **Preprocessing:** Improves image quality by removing noise.
3. **Feature Extraction:** Identifies important features such as edges, shapes and colours.
4. **Analysis:** Uses algorithms or AI models to understand the image.
5. **Output:** Produces meaningful results such as object identification or classification.

Conclusion: Computer Vision helps machines understand visual information and is widely used in healthcare, robotics, security, transportation and automation.

Unit 1 • Q2

Explain Geometric Primitives and Discuss Their Importance in Image Analysis.

Exam Question: 2. Explain Geometric primitives and discuss their importance in image analysis

FORMULA **2D Homogeneous Point and Line Incidence Condition**

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad \mathbf{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \mathbf{x}^T\mathbf{l} = ax + by + c = 0$$

In homogeneous coordinates, point x lies on line l if and only if their inner product vanishes.

FORMULA **3D Homogeneous Plane and Point Incidence Condition**

$$\pi = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \quad \pi^T\mathbf{X} = aX + bY + cZ + d = 0$$

Represents a 3D planar primitive and the condition for a 3D scene point to lie on the plane.

Unit 1 – Q2. Explain Geometric Primitives and Discuss Their Importance in Image Analysis.

1. Definition of Geometric Primitives

Geometric Primitives are basic geometric shapes or elements used to represent objects and their structures in an image. They are the fundamental building blocks of image analysis.

In Computer Vision, geometric primitives help computers identify object shapes, boundaries, positions and spatial relationships.

2. Types of Geometric Primitives

1. **Point:** Represents a specific location in an image using coordinates (x, y). It is used for feature detection and tracking.
2. **Line:** Represents a straight connection between two points. It helps detect object boundaries and structural patterns.
3. **Edge:** Represents a boundary where image intensity or colour changes suddenly. It helps identify object shapes.
4. **Circle:** A closed geometric shape defined by its centre and radius. It is used in detecting circular objects.
5. **Ellipse:** An oval-shaped structure used to represent curved objects and regions.
6. **Polygon:** A closed shape made of multiple connected line segments. It helps represent complex object boundaries.
7. **Rectangle:** A four-sided shape used to define object regions and bounding boxes.

3. Importance in Image Analysis

1. **Shape Detection:** Helps identify different objects based on their geometric shapes.

2. **Object Recognition:** Helps recognize objects using their boundaries and structural features.
3. **Image Segmentation:** Helps divide images into meaningful regions.
4. **Feature Extraction:** Extracts important information such as points, edges and lines.
5. **Object Tracking:** Helps track object positions and movements across video frames.
6. **3D Reconstruction:** Helps estimate the three-dimensional structure of objects from multiple images.

4. Example

In traffic sign recognition, geometric primitives help identify signs:

- Circle – Speed limit signs.
- Triangle – Warning signs.
- Rectangle – Information signs.

Conclusion: Geometric primitives provide a simple mathematical representation of image structures and are essential for object detection, recognition, segmentation and image understanding.

Unit 1 • Q3

Explain Photometric Image Formation and Digital Camera Model with Neat Diagrams.

Exam Question: 3. Explain photometric image formation and digital camera model with neat diagrams.

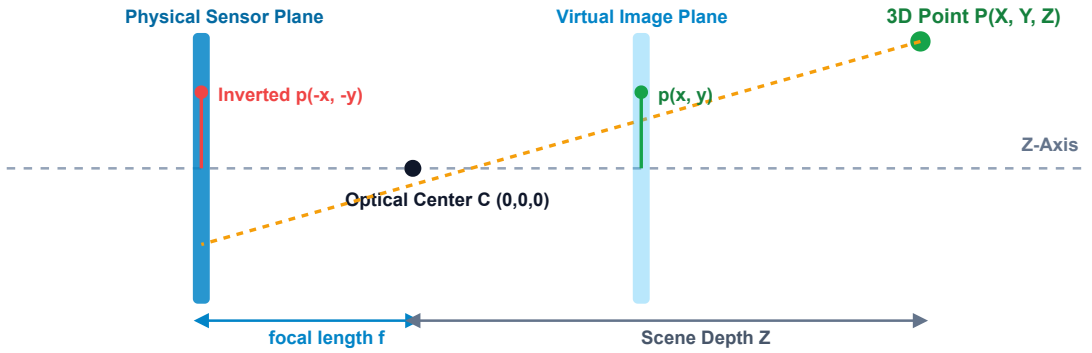


FIGURE 1.1: Pinhole Camera Geometry and Perspective Projection Model

Light ray passing through optical center C maps 3D point P(X,Y,Z) to image coordinates via similar triangles: x = f·(X/Z), y = f·(Y/Z).



FIGURE 1.2: Photometric Image Formation & Digital Camera Pipeline

Continuous optical irradiance converts to digital discrete pixel matrix via optics, sensor exposure, and A/D sampling.

FORMULA **Ideal Perspective Projection Equations**

$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

Maps 3D real-world coordinates (X, Y, Z) to 2D image coordinates (x, y) with focal length f.

FORMULA **Full Digital Camera Projection Matrix**

$$\mathbf{x} = \mathbf{K}[\mathbf{R} \mid \mathbf{t}]\mathbf{X}, \quad \mathbf{K} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Combines intrinsic calibration matrix K with extrinsic rotation R and translation vector t.

Unit 1 – Q3. Explain Photometric Image Formation and Digital Camera Model with Neat Diagrams.

1. Photometric Image Formation

Photometric Image Formation explains how light interacts with an object and how a camera captures that light to form an image.

The appearance of an object depends on three main factors:

1. **Light Source:** Provides illumination to the object.
2. **Object Surface:** Reflects or absorbs the incoming light depending on its material and texture.
3. **Camera Sensor:** Receives the reflected light and converts it into image information.

Basic Image Formation Diagram:

Important factors affecting image formation:

- **Illumination:** Amount of light falling on an object.
- **Reflectance:** Ability of an object to reflect light.
- **Surface Orientation:** Direction of the object surface affects reflected light.
- **Shading:** Variation in brightness due to light and surface shape.

A simple Lambertian reflection model is:

[I = \rho E \max(0,\cos\theta)]

Where:

- (I) = Image intensity
- (\rho) = Surface reflectance
- (E) = Light intensity
- (\theta) = Angle between surface normal and light direction.

2. Digital Camera Model

A **Digital Camera Model** explains how a real-world scene is converted into a digital image using a camera lens and electronic sensor.

It consists of the following steps:

1. **Light Collection:** The camera lens collects light reflected from objects.
2. **Image Projection:** The lens focuses light onto the image sensor.
3. **Sensing:** The sensor (CCD or CMOS) converts light into electrical signals.
4. **Digitization:** Signals are converted into digital pixel values using an Analog-to-Digital Converter (ADC).
5. **Image Storage:** The digital image is stored as a matrix of pixels.

Digital Camera Model Diagram:

3. Mathematical Representation

The digital image can be represented as:

[I(x,y)]

Where: - (x,y) = Pixel coordinates. - (I(x,y)) = Intensity value of the pixel.

For an 8-bit grayscale image, intensity ranges from **0 to 255**.

Conclusion

Photometric image formation explains how light creates an image, while the digital camera model explains how the camera captures and converts that light into digital pixels. Both are fundamental concepts in Computer Vision.

Unit 1 • Q4

Analyze the Role of Fourier Transform in Image Processing and Justify Its Importance.

> 📝 **Exam Question:** 4. Analyze the role of Fourier Transform in image processing and justify its importance.

Unit 1 – Q4. Analyze the Role of Fourier Transform in Image Processing and Justify Its Importance.

1. Introduction

Fourier Transform (FT) is a mathematical technique used in image processing to convert an image from the **Spatial Domain** into the **Frequency Domain**.

It represents an image as a combination of different frequency components. It helps identify patterns, edges, noise and repeated structures in an image.

2. Types of Frequencies in Images

1. **Low Frequency:** Represents slowly changing intensity, smooth areas and background information.
2. **High Frequency:** Represents rapid intensity changes, edges, fine details and noise.

3. Fourier Transform Formula

The 2D Discrete Fourier Transform (DFT) is:

[F(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}]

Where:

- (f(x,y)) = Original image.
- (F(u,v)) = Frequency representation.
- (M,N) = Image dimensions.
- (u,v) = Frequency coordinates.
- (j) = Imaginary unit.

4. Working of Fourier Transform

The working process involves:

1. **Input Image:** Take a digital image as input.
2. **Transformation:** Apply Fourier Transform to convert it into frequency components.
3. **Frequency Analysis:** Identify low-frequency and high-frequency information.
4. **Filtering:** Remove unwanted frequencies or enhance important frequencies.
5. **Inverse Fourier Transform:** Convert the processed frequency information back into an image.

5. Applications of Fourier Transform

1. **Image Filtering:** Used for low-pass and high-pass filtering.
2. **Noise Reduction:** Removes unwanted frequency components.
3. **Image Enhancement:** Improves image details and quality.
4. **Image Compression:** Helps represent image information efficiently.

5. **Edge Detection:** Highlights sudden intensity changes.
6. **Pattern Recognition:** Helps identify repeated patterns and textures.

6. Importance of Fourier Transform

- Makes frequency analysis easier.
- Helps separate noise from useful image information.
- Simplifies filtering operations.
- Supports image restoration and enhancement.
- Helps analyze periodic patterns in images.

Conclusion

Fourier Transform is important in image processing because it converts images into the frequency domain, making filtering, noise removal, enhancement and pattern analysis easier and more effective.

Unit 1 • Q5

Explain the 2D Discrete Fourier Transform (DFT) and Its Significance in Image Processing.

📝 **Exam Question:** 5. Explain the 2D Discrete Fourier Transform (DFT) and its significance in image processing.

FORMULA **2D Discrete Fourier Transform (DFT)**

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

Converts spatial pixel intensities into orthogonal spatial frequency components (u, v).

FORMULA **2D Inverse Discrete Fourier Transform (IDFT)**

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

Reconstructs spatial domain image f(x,y) from frequency magnitude and phase spectra.

Unit 1 – Q5. Explain the 2D Discrete Fourier Transform (DFT) and Its Significance in Image Processing.

1. Definition of 2D Discrete Fourier Transform (DFT)

The **2D Discrete Fourier Transform (DFT)** is a mathematical technique used to convert a digital image from the **Spatial Domain** into the **Frequency Domain**.

It represents an image as a combination of different frequency components. It helps analyze image patterns, edges, textures and noise.

2. Mathematical Formula

The 2D DFT of an image is given by:

[F(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(\frac{ux}{M}+\frac{vy}{N})}]

Where:

- (f(x,y)) = Input image.
- (F(u,v)) = Frequency domain representation.
- (M,N) = Image dimensions.
- (x,y) = Spatial coordinates.
- (u,v) = Frequency coordinates.
- (j) = Imaginary unit.

3. Working of 2D DFT

The working process involves the following steps:

1. **Input Image:** Take a digital image represented as a pixel matrix.
2. **DFT Application:** Convert the image into frequency components.
3. **Frequency Analysis:** Identify low and high-frequency components.

4. **Frequency Filtering:** Remove unwanted frequencies or enhance important ones.
5. **Inverse DFT (IDFT):** Convert the processed frequency data back into the spatial domain.

4. Types of Frequency Components

1. **Low Frequency:** Represents smooth areas and gradual intensity changes.
2. **High Frequency:** Represents edges, sharp details and noise.

5. Significance in Image Processing

1. **Image Filtering:** Performs low-pass and high-pass filtering.
2. **Noise Removal:** Helps eliminate unwanted noise.
3. **Image Enhancement:** Improves image quality and details.
4. **Image Compression:** Helps in frequency-based image representation.
5. **Edge Detection:** Helps identify sharp changes in intensity.
6. **Pattern Recognition:** Helps analyze repeated patterns and textures.

6. Inverse DFT Formula

[f(x,y)=\frac{1}{MN}\sum_{u=0}^{M-1} \sum_{v=0}^{N-1}F(u,v)e^{j2\pi(\frac{ux}{M}+\frac{vy}{N})}]

It converts the frequency-domain representation back into the original spatial-domain representation.

Conclusion: 2D DFT is an important image processing technique because it helps analyze frequency information and perform filtering, noise reduction, enhancement and image restoration efficiently.

Unit 1 • Q6

Explain Geometric Transformations (Orthogonal, Euclidean, Affine and Projective).

Exam Question: 6. Explain Geometric Transformations (orthogonal, Euclidean, Affine, projective).



FIGURE 1.3: **Hierarchy of 2D Planar Geometric Transformations**

Degrees of freedom increase from Euclidean (3 DOF) to Projective (8 DOF), progressively relaxing geometric invariants.

FORMULA **Planar Projective Transformation (Homography - 8 DOF)**

$$\mathbf{x} = \mathbf{Hx} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \mathbf{x}$$

Most general planar linear transformation preserving collinearity and cross-ratios.

Unit 1 – Q6. Explain Geometric Transformations (Orthogonal, Euclidean, Affine and Projective).

1. Definition of Geometric Transformations

Geometric Transformation is the process of changing the position, size, shape or orientation of an image or object using mathematical operations.

In Computer Vision, geometric transformations are used for image alignment, image resizing, rotation, object tracking and image stitching.

2. Types of Geometric Transformations

1. Orthogonal Transformation

Orthogonal transformation preserves distances and angles. It includes rotation and reflection.

- Maintains the shape and size of objects.
- Preserves perpendicular relationships.
- Used in image orientation and alignment.

2. Euclidean Transformation

Euclidean transformation includes rotation and translation.

- Preserves distances and angles.
- Maintains the original shape and size.

- Changes the position and orientation of objects.

Formula:

[\begin{bmatrix}x'\y'\end{bmatrix} = \begin{bmatrix} \cos\theta&-\sin\theta\ \sin\theta&\cos\theta \end{bmatrix} \begin{bmatrix}x\y\end{bmatrix} + \begin{bmatrix}t_x\t_y\end{bmatrix}]

Where (\theta) is the rotation angle and (t_x,t_y) are translation values.

3. Affine Transformation

Affine transformation preserves straight lines and parallelism but may change angles and distances.

- Includes rotation, translation, scaling and shearing.
- Preserves parallel lines.
- Used in image resizing and image alignment.

Formula:

[\begin{bmatrix}x'\y'\end{bmatrix} = \begin{bmatrix}a&b\c&d\end{bmatrix} \begin{bmatrix}x\y\end{bmatrix} + \begin{bmatrix}t_x\t_y\end{bmatrix}]

4. Projective Transformation

Projective transformation changes the perspective of an image. It preserves straight lines but does not necessarily preserve parallelism, angles or distances.

- Also called **Homography**.
- Represents perspective changes.
- Used in panorama stitching and perspective correction.

Formula:

[\begin{bmatrix}x'\y'\w'\end{bmatrix} = H\begin{bmatrix}x\y\1\end{bmatrix}]

Where (H) is a 3×3 transformation matrix.

3. Comparison of Transformations

Transformation	Preserves
Orthogonal	Distance and angles
Euclidean	Distance and angles
Affine	Straight lines and parallelism
Projective	Straight lines

4. Applications

1. Image alignment and registration.
2. Image stitching and panorama creation.
3. Object recognition and tracking.
4. Perspective correction.
5. Image resizing and rotation.

Conclusion: Geometric transformations are important in Computer Vision because they help change image coordinates and align images while maintaining specific geometric properties.

Unit 1 • Q7

Describe Point Operators, Linear Filtering and Global Optimization Methods with Suitable Examples.

> 📝 **Exam Question:** 7. Describe point operators, linear filtering and Global optimization methods suitable examples.

FORMULA **Markov Random Field (MRF) Energy Minimization**

$$E(f) = \sum_{p \in V} D_p(f_p) + \lambda \sum_{(p,q) \in \mathcal{N}} V_{p,q}(f_p, f_q)$$

Balances data fidelity D_p (likelihood) with spatial smoothness V_{p,q} (prior penalty).

Unit 1 – Q7. Describe Point Operators, Linear Filtering and Global Optimization Methods with Suitable Examples.

Image processing uses different techniques to improve image quality, remove noise and extract useful information. Three important methods are **Point Operators, Linear Filtering and Global Optimization**.

1. Point Operators

Point Operators are image processing techniques in which each output pixel depends only on the corresponding input pixel value.

They do not depend on neighbouring pixels.

Formula:

[g(x,y)=T(f(x,y))]

Where: - (f(x,y)) = Input pixel value. - (g(x,y)) = Output pixel value. - (T) = Transformation function.

Examples: 1. Brightness adjustment. 2. Contrast enhancement. 3. Image thresholding. 4. Negative image transformation.

Example: If the original pixel value is 100 and brightness is increased by 50, the new pixel value becomes 150.

2. Linear Filtering

Linear Filtering is a technique in which each output pixel is calculated using a weighted combination of neighbouring input pixels.

It uses a small matrix called a **Filter Kernel** or **Mask**.

Formula:

[g(x,y)=\sum_i\sum_j w(i,j)f(x-i,y-j)]

Where: - (f) = Input image. - (g) = Output image. - (w) = Filter kernel.

Examples:

1. **Mean Filter:** Reduces noise by averaging neighbouring pixels.
2. **Gaussian Filter:** Smooths images and reduces noise.
3. **Laplacian Filter:** Enhances edges and image details.
4. **Sobel Filter:** Detects horizontal and vertical edges.

3. Global Optimization Methods

Global Optimization is a technique that finds the best overall solution by minimizing or maximizing an objective function over the entire image or problem.

It considers relationships between multiple pixels instead of processing each pixel independently.

Examples:

1. **Graph Cut:** Divides an image into foreground and background regions.
2. **Normalized Cuts:** Segments an image using graph-based similarity.
3. **Dynamic Programming:** Finds an optimal solution by solving smaller subproblems.
4. **Energy Minimization:** Finds the best image labels by minimizing an energy function.

Example: Graph Cut can separate a person from the background in an image.

4. Comparison

Method	Main Purpose
Point Operators	Change individual pixel values
Linear Filtering	Process neighbouring pixels
Global Optimization	Find the best overall solution

Conclusion: Point operators modify individual pixels, linear filtering processes neighbouring pixels, and global optimization considers overall image relationships. These methods are important for image enhancement, noise removal and segmentation.

Unit 1 • Q8

Apply Histogram Equalization Technique to Enhance an Image and Explain the Steps.

Exam Question: 8. Apply histogram equalization technique to enhance an image and explain the steps.

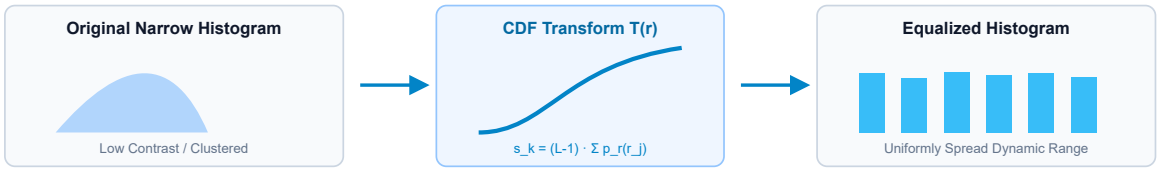


FIGURE 1.4: Histogram Equalization Transformation Process

Using the Cumulative Distribution Function (CDF) as a monotonic transfer function spreads pixel intensities across [0, L-1].

FORMULA **Histogram Equalization Transformation Function**

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

Maps input gray level r_k to equalized level s_k using the discrete Cumulative Distribution Function.

Unit 1 – Q8. Apply Histogram Equalization Technique to Enhance an Image and Explain the Steps.

1. Definition of Histogram Equalization

Histogram Equalization is an image enhancement technique used to improve the contrast of an image by distributing its intensity values more uniformly.

It is especially useful for images that appear too dark or have low contrast. It improves visibility by spreading pixel intensity values across the available intensity range.

2. Steps of Histogram Equalization

Step 1: Read the Input Image

Take a grayscale image as input. For an 8-bit image, pixel intensity ranges from 0 to 255.

Step 2: Calculate Histogram

Count the number of pixels for each intensity level.

Step 3: Calculate Probability

Calculate the probability of each intensity value:

[P(r_k)=\frac{n_k}{N}]

Where: - (n_k) = Number of pixels with intensity (r_k). - (N) = Total number of pixels.

Step 4: Calculate Cumulative Distribution Function (CDF)

Calculate the cumulative sum of probabilities.

[CDF(r_k)=\sum_{j=0}^{k}P(r_j)]

Step 5: Calculate New Intensity Values

Apply the histogram equalization formula:

[s_k=(L-1)\times CDF(r_k)]

Where: - (L) = Total number of intensity levels. - (s_k) = New intensity value.

Step 6: Generate Enhanced Image

Replace the original pixel values with the calculated new intensity values to obtain an enhanced image.

3. Simple Example

Suppose an 8-bit grayscale image has:

- Minimum intensity = 50
- Maximum intensity = 150
- Total intensity levels = 256

Histogram equalization redistributes the intensity values across a wider range, improving image contrast.

4. Applications

1. Medical image enhancement.
2. Satellite image processing.
3. Low-light image improvement.
4. Image analysis and object detection.
5. Improving image visibility.

5. Advantages

- Improves image contrast.
- Highlights hidden details.
- Simple and computationally efficient.
- Useful for low-contrast images.

Conclusion: Histogram Equalization improves image quality by redistributing pixel intensity values using the cumulative distribution function (CDF), making important image details more visible.

Unit 1 • Q9

Describe Image Restoration Techniques and Explain How Global Optimization Methods Are Used in Computer Vision.

📝 **Exam Question:** 9. Describe image restoration techniques and explain how global optimization methods are used in computer vision.

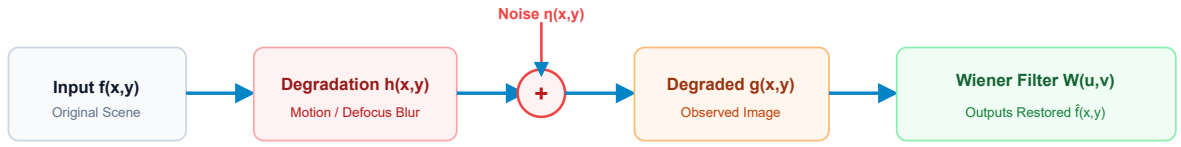


FIGURE 1.5: Image Degradation and Restoration Model

*Model: g(x,y) = f(x,y)*h(x,y) + η(x,y). Optimal Wiener restoration minimizes mean square error E{(f - f̂)²}.*

FORMULA Optimal Wiener Restoration Filter Transfer Function

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \frac{S_\eta(u,v)}{S_f(u,v)}} = \frac{H^*(u,v)}{|H(u,v)|^2 + K}$$

Minimizes mean square error between original image and restored estimate in presence of blur H and noise.

Unit 1 – Q9. Describe Image Restoration Techniques and Explain How Global Optimization Methods Are Used in Computer Vision.

1. Image Restoration

Image Restoration is the process of recovering an original image from a degraded or damaged image. It removes unwanted effects such as noise, blur, distortion and other image defects.

It uses mathematical models and algorithms to improve image quality.

2. Image Restoration Techniques

1. Noise Removal

Removes unwanted random variations from an image.

- Example: Median Filter and Gaussian Filter.
- Used to improve image clarity.

2. Image Deblurring

Removes blur caused by camera movement or incorrect focusing.

- Example: Wiener Filtering.
- Helps recover sharp image details.

3. Inverse Filtering

Uses the degradation model to recover the original image.

Formula:

[F(u,v)=\frac{G(u,v)}{H(u,v)}]

Where: - (G(u,v)) = Degraded image. - (H(u,v)) = Degradation function. - (F(u,v)) = Restored image.

4. Wiener Filtering

Reduces noise and blur by considering both image degradation and noise.

5. Inpainting

Reconstructs missing or damaged parts of an image using surrounding information.

3. Global Optimization Methods in Computer Vision

Global Optimization is a technique used to find the best overall solution by minimizing or maximizing an objective function.

In Computer Vision, it considers relationships between multiple pixels or image features to produce accurate results.

4. Methods of Global Optimization

1. **Graph Cut:** Divides an image into foreground and background by finding a minimum cut in a graph.
2. **Normalized Cuts:** Segments an image into meaningful regions based on similarity between pixels.
3. **Energy Minimization:** Finds the best solution by minimizing an energy function.
4. **Dynamic Programming:** Breaks a complex problem into smaller subproblems to find an optimal solution.

5. Applications

- Image segmentation.
- Object recognition.
- Noise removal.
- 3D reconstruction.
- Image matching and alignment.
- Motion estimation.

6. Importance

- Improves image quality.
- Provides accurate segmentation.
- Reduces image noise and errors.
- Finds optimal solutions by considering global relationships.

Conclusion: Image restoration recovers damaged images, while global optimization finds the best overall solution in computer vision. Both are important for accurate image analysis and processing.

Unit 1 • Q10

What Are the Different Types of Filters? Explain with Examples.

> 📝 **Exam Question:** 10. What are the different types of filters. Explain with example.

Unit 1 – Q10. What Are the Different Types of Filters? Explain with Examples.

1. Definition of Filters

Filters are image processing techniques used to modify or improve an image by removing noise, smoothing regions, sharpening details and detecting edges.

Filters operate on pixel values using mathematical operations. They are mainly classified into the following types.

2. Different Types of Filters

1. Low-Pass Filter (Smoothing Filter)

A Low-Pass Filter allows low-frequency components and reduces high-frequency components.

- Removes noise.
- Smooths the image.
- Reduces small details.

Example: Gaussian Filter and Mean Filter.

2. High-Pass Filter (Sharpening Filter)

A High-Pass Filter preserves high-frequency components and reduces low-frequency components.

- Highlights edges.
- Enhances fine details.
- Improves image sharpness.

Example: Laplacian Filter and High-Pass Kernel.

3. Median Filter

Median Filter is a non-linear filter that replaces each pixel with the median value of its neighbouring pixels.

- Removes salt-and-pepper noise.
- Preserves edges better than mean filtering.

Example: Removing black and white noise from an image.

4. Gaussian Filter

Gaussian Filter uses a Gaussian function to calculate weighted averages of neighbouring pixels.

- Reduces image noise.
- Produces smooth images.
- Used in preprocessing.

Example: Blurring an image before edge detection.

5. Mean Filter

Mean Filter replaces each pixel with the average value of its neighbouring pixels.

- Reduces random noise.
- Smooths image variations.

Example: Applying a 3×3 averaging kernel.

6. Edge Detection Filter

These filters identify boundaries where image intensity changes suddenly.

- Detects object boundaries.
- Extracts important image features.

Example: Sobel Filter and Prewitt Filter.

3. Comparison of Filters

Filter	Main Purpose
Low-Pass	Noise reduction
High-Pass	Image sharpening
Median	Salt-and-pepper noise removal
Gaussian	Image smoothing
Mean	Averaging neighbouring pixels
Sobel	Edge detection

4. Applications

1. Medical image processing.
2. Object detection and recognition.
3. Image enhancement.
4. Noise removal.
5. Edge detection and segmentation.

Conclusion: Filters are essential in Computer Vision because they improve image quality, remove noise, sharpen details and extract important features for further image analysis.

QUICK REVISION / COMPLETE RECAP

Unit 1: Fundamentals of Image Formation & Enhancement — Complete Recap

High-yield revision summary, definitions, key points to remember, and core formulas for last-minute exam prep.

1. Unit Overview & Summary

Unit 1 establishes the mathematical foundations of Computer Vision, covering how optical sensors capture real-world scenes, how geometry governs spatial mapping, and how mathematical transformations and filters enhance and restore degraded digital images.

2. Core Concepts & Definitions

- **Computer Vision (CV)**: The multidisciplinary field of Artificial Intelligence enabling computing systems to acquire, process, analyze, and comprehend high-dimensional visual data from the physical world.
- **Pinhole Camera Model**: An idealized geometric projection model where light rays pass through an infinitesimally small aperture (pinhole / optical center) onto an image plane, mapping 3D scene point (X, Y, Z) to 2D image coordinates (x, y) via perspective projection:

  FORMULA Governing Mathematical Equation

  $$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

- **Geometric Primitives**:
- **Points**: 2D points $(x, y) \in \mathbb{R}^2$ represented in homogeneous coordinates as $\tilde{\mathbf{x}} = (x, y, 1)^T$. 3D points (X, Y, Z) represented as $(X, Y, Z, 1)^T$.
- **Lines**: In 2D, lines represented as $\mathbf{l} = (a, b, c)^T$ such that $\mathbf{x}^T \mathbf{l} = ax + by + c = 0$.
- **Planes**: In 3D, represented as $\mathbf{\pi} = (a, b, c, d)^T$ where $aX + bY + cZ + d = 0$.
- **Hierarchy of 2D Geometric Transformations**:
- **Euclidean (Rigid)** (3 DOF): Preserves Euclidean distance, angles, and shapes. Rotation + Translation.

  FORMULA Governing Mathematical Equation

  $$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2. **Similarity** (4 DOF): Rigid transformation + isotropic scale factor s. Preserves angles and aspect ratio. 3. **Affine** (6 DOF): Preserves parallel lines and ratios of lengths along parallel lines. Rectangles become parallelograms. 4. **Projective (Homography)** (8 DOF): Most general planar linear transformation. Preserves straight lines and cross-ratios. - **2D Discrete Fourier Transform (DFT)**: Transforms a spatial domain image $f(x,y)$ of size $M \times N$ into frequency components $F(u,v)$:

FORMULA **Governing Mathematical Equation**

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

- **Low frequencies**: Represent smooth regions, illumination, and background.
- **High frequencies**: Represent sharp transitions, edges, textures, and fine noise.

- **Histogram Equalization**: A non-linear point operator that spreads pixel intensities evenly across the full dynamic range $[0, L-1]$ by using the Cumulative Distribution Function (CDF):

FORMULA **Governing Mathematical Equation**

$$s_k = T(r_k) = (L-1) \sum_{j=0}^{k} p_r(r_j) = \frac{L-1}{MN} \sum_{j=0}^{k} n_j$$

- **Image Restoration vs Enhancement**:

- *Enhancement*: Subjective, heuristic technique to improve visual appeal for human eyes (e.g., contrast stretching, unsharp masking).
- *Restoration*: Objective, mathematical inversion process to recover an uncorrupted image using an explicit mathematical degradation model:

FORMULA **Governing Mathematical Equation**

$$g(x,y) = f(x,y) * h(x,y) + \eta(x,y) \Leftrightarrow G(u,v) = F(u,v)H(u,v) + N(u,v)$$

- **Wiener Filter**: The optimal Mean Square Error (MSE) linear restoration filter balancing inverse filtering and noise suppression:

FORMULA **Governing Mathematical Equation**

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \frac{S_\eta(u,v)}{S_f(u,v)}} = \frac{H^*(u,v)}{|H(u,v)|^2 + K}$$

- **Markov Random Fields (MRF)**: A probabilistic framework modeling spatial dependencies between neighboring pixels, formulated as Gibbs energy minimization:

FORMULA **Governing Mathematical Equation**

$$E(f) = E_{\text{data}}(f) + \lambda E_{\text{smooth}}(f)$$

---

3. Key Points to Remember for Exams

1. Perspective projection introduces scale ambiguity: objects farther from the camera appear smaller by factor $1/Z$.

2. In homogeneous coordinates, points at infinity have scale coordinate $w = 0$, allowing parallel lines to intersect at vanishing points.
3. Histogram equalization produces a uniform distribution in continuous theory, but discrete images produce a flattened, spread-out histogram with gaps.
4. Direct Inverse Filtering ($1/H$) amplifies noise catastrophically at frequencies where $H(u,v) \approx 0$. Wiener filtering solves this by adding the noise-to-signal power ratio K.
5. Spatial convolution in the spatial domain corresponds to point-wise multiplication in the frequency domain ($f * h \iff F \cdot H$).

---

4. Canonical Algorithms & Formulas Summary

- **Pinhole Camera Matrix**: $\mathbf{P} = \mathbf{K} [\mathbf{R} \mid \mathbf{t}]$, where $\mathbf{K}$ is internal calibration matrix:

FORMULA **Governing Mathematical Equation**

$$\mathbf{K} = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

- **Linear Filters**:

- *Mean / Box Filter*: Low-pass spatial smoothing; blurs edges.
- *Gaussian Filter*: Weighted isotropic smoothing; preserves edges better than mean filter.
- *Median Filter*: Non-linear filter; exceptional at removing salt-and-pepper noise without blurring sharp edges.
- *Sobel Operator*: First-order directional edge detector (G_x, G_y).
- *Laplacian Operator*: Second-order isotropic derivative detecting zero-crossings (${\nabla}^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$).

✎ SHORT ANSWER QUESTIONS (2 MARKS)

Unit 1 — 2-Mark Exam Questions

Crisp, direct, point-to-point answers designed for maximum marks in university short-answer sections.

Q1 What is Computer Vision?

Computer Vision is a branch of Artificial Intelligence (AI) that enables computers to understand, analyze and interpret images and videos. It helps machines identify objects, recognize faces, detect movements and understand scenes. Examples include face recognition, self-driving cars and medical image analysis.

Q2 What is Image Enhancement?

Image Enhancement is the process of improving the visual quality of an image to make important details clearer. It includes techniques such as contrast adjustment, brightness correction, sharpening and noise removal. It helps improve image visibility and makes further image processing easier.

Q3 What is Projective Transformation in Computer Vision?

Projective Transformation is a geometric transformation that changes the perspective of an image while preserving straight lines. It is commonly used to correct image perspective and align images. It is represented using a **3×3 Homography Matrix** and is useful in panorama stitching.

Q4 Write the Applications of Filtering.

Filtering is used in image processing to improve image quality and extract useful information. Its main applications are: 1. **Noise Removal:** Removes unwanted noise. 2. **Image Sharpening:** Enhances edges and details. 3. **Smoothing:** Reduces image variations. 4. **Edge Detection:** Identifies object boundaries.

Q5 What is Image Restoration?

Image Restoration is the process of recovering an original image from a degraded or damaged image. It removes unwanted effects such as noise, blur and distortion using mathematical techniques. Common methods include inverse filtering, Wiener filtering and deblurring.

UNIT 2

Unit 2: Image Segmentation & Feature Extraction

Comprehensive Question & Answer Collection • Quick Revision Recap • 2-Mark Direct Questions

PART A: DETAILED EXAMINATION QUESTIONS & ANSWERS

Unit 2 • Q1

Evaluate Segmentation Techniques Such as Normalized Cut and Graph Cut.

📝 **Exam Question:** **Unit-II** 1\. Evaluate segmentation techniques such as normalized cut and graph cut.

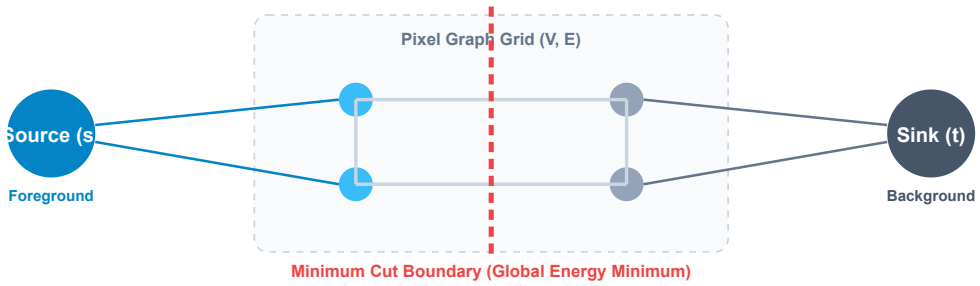


FIGURE 2.1: Graph Cut Segmentation Model (s-t Min-Cut Max-Flow)

Boykov-Kolmogorov graph cut partitions pixels into foreground/background by severing minimum weight capacity edges.

FORMULA **Normalized Cut (N-cut) Objective Function**

$$\text{Ncut}(A,B) = \frac{\text{cut}(A,B)}{\text{assoc}(A,V)} + \frac{\text{cut}(A,B)}{\text{assoc}(B,V)}$$

Overcomes small-cluster bias of standard min-cut by penalizing unbalanced partitions.

FORMULA **Normalized Cut Generalized Eigenvalue Problem**

$$(D-W)y = \lambda Dy$$

Relaxed continuous solver where D is degree matrix, W is affinity weight matrix, and y is indicator vector.

Unit II – Q1. Evaluate Segmentation Techniques Such as Normalized Cut and Graph Cut.

1. Introduction to Image Segmentation

Image Segmentation is the process of dividing an image into multiple meaningful regions based on similarities in colour, intensity, texture or other features.

Two important graph-based segmentation techniques are **Normalized Cut** and **Graph Cut**.

2. Normalized Cut (N-Cut)

Normalized Cut is a graph-based image segmentation technique that divides an image into meaningful regions by minimizing the similarity between different groups while maintaining similarity within each group.

Working Steps:

1. **Graph Construction:** Represent image pixels as graph nodes.
2. **Edge Formation:** Connect neighbouring pixels using weighted edges based on similarity.
3. **Similarity Calculation:** Calculate the similarity between pixels.

4. **Graph Partitioning:** Divide the graph into different regions.
5. **Segmentation:** Produce the final segmented image.

Formula:

[Ncut(A,B)=\frac{cut(A,B)}{assoc(A,V)} +\frac{cut(A,B)}{assoc(B,V)}]

Where: - (A,B) = Two image regions. - (cut(A,B)) = Connection weight between regions. - (assoc(A,V)) = Total connection weight of region A to the graph.

Example: Separating different objects in an image based on colour and similarity.

3. Graph Cut

Graph Cut is an image segmentation technique that separates foreground and background by finding a minimum cut in a graph.

Working Steps:

1. **Graph Construction:** Represent pixels as nodes.
2. **Terminal Nodes:** Create foreground and background terminal nodes.
3. **Edge Weights:** Assign weights based on pixel similarity.
4. **Minimum Cut:** Find the minimum-cost cut that separates the graph.
5. **Segmentation:** Classify pixels into foreground and background.

Example: Separating a person from the background in a photograph.

4. Comparison Between Normalized Cut and Graph Cut

Feature	Normalized Cut	Graph Cut
Approach	Graph partitioning	Minimum cut
Main purpose	Divide image into regions	Separate foreground and background
Basis	Similarity between pixels	Energy minimization
Output	Multiple meaningful regions	Foreground and background
Complexity	Computationally expensive	Depends on graph and energy model

5. Applications

1. Medical image segmentation.
2. Object detection and recognition.
3. Background removal.
4. Image analysis.
5. Video processing.

6. Evaluation

- **Normalized Cut:** Useful for dividing images into multiple meaningful regions, but can be computationally expensive for large images.
- **Graph Cut:** Effective for foreground-background separation and energy minimization, but its result depends on the chosen energy model and user-provided constraints.

Conclusion: Both Normalized Cut and Graph Cut are important graph-based segmentation techniques. Normalized Cut focuses on dividing images into similar regions, whereas Graph Cut focuses on finding an optimal separation between regions.

Unit 2 • Q2

List and Explain the Different Types of Feature Detectors Used in Computer Vision.

📝 **Exam Question:** 2. List and explain the different types of feature detectors used in computer vision.

FORMULA **Harris Corner Detector Structure Tensor & Response Function**

$$\mathbf{M} = \sum_{(x,y) \in W} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \quad R = \det(\mathbf{M}) - k(\text{trace}(\mathbf{M}))^2$$

M measures local intensity variance; R > 0 signifies a corner, R < 0 an edge, and |R| ≈ 0 a flat region.

Unit II – Q2. List and Explain the Different Types of Feature Detectors Used in Computer Vision.

1. Definition of Feature Detection

Feature Detection is the process of identifying important points, edges, corners and regions in an image. These features help computers understand image structures and recognize objects.

Feature detectors are widely used in object recognition, image matching, tracking and 3D reconstruction.

2. Different Types of Feature Detectors

1. Harris Corner Detector

- Detects corners where image intensity changes significantly in two directions.
- Identifies important corner points.
- Used in image matching and object tracking.

Example: Detecting corners of a building.

2. SIFT (Scale-Invariant Feature Transform)

- Detects keypoints at different image scales.
- Uses Difference of Gaussian (DoG) for feature detection.
- Works under changes in scale and rotation.
- Used in object recognition and image stitching.

Example: Matching the same object in differently sized images.

3. SURF (Speeded-Up Robust Features)

- Uses Hessian Matrix to detect keypoints.
- Uses Integral Images for faster computation.
- Provides scale and rotation robustness.
- Used in image matching and recognition.

Example: Identifying objects from different viewpoints.

4. FAST (Features from Accelerated Segment Test)

- Detects corner points by comparing surrounding pixels.
- Very fast and computationally efficient.
- Commonly used in real-time applications.

Example: Corner detection in mobile camera applications.

5. ORB (Oriented FAST and Rotated BRIEF)

- Combines FAST keypoint detection with BRIEF descriptors.
- Provides rotation robustness.
- Faster and more memory-efficient than SIFT.
- Used in real-time object tracking and image matching.

6. SUSAN Detector

- Detects corners and edges by comparing brightness values of neighbouring pixels.
- Does not require image gradients.
- Used in feature extraction and image analysis.

3. Feature Detector Diagram

4. Comparison of Feature Detectors

Detector	Main Feature
Harris	Corner detection
SIFT	Scale and rotation invariant keypoints
SURF	Fast robust feature detection
FAST	High-speed corner detection
ORB	Fast oriented feature detection
SUSAN	Corner and edge detection

5. Applications

1. Object recognition.
2. Image stitching.
3. Object tracking.
4. 3D reconstruction.
5. Image matching.
6. Motion estimation.

Conclusion: Feature detectors identify important image structures such as corners, edges and keypoints. SIFT, SURF, Harris, FAST and ORB are widely used techniques that help improve computer vision applications.

Unit 2 • Q3

Demonstrate the Feature Matching Process Using Feature Descriptors.

> 📝 **Exam Question:** 3. Demonstrate the feature matching process using feature descriptors.

Unit II – Q3. Demonstrate the Feature Matching Process Using Feature Descriptors.

1. Definition of Feature Matching

Feature Matching is the process of identifying and matching similar features between two or more images. It uses feature descriptors to compare important points such as corners, edges and keypoints.

It is commonly used in image stitching, object recognition, motion tracking and 3D reconstruction.

2. What are Feature Descriptors?

Feature Descriptors are numerical representations of detected keypoints. They describe the local appearance of an image around a particular feature.

Common feature descriptors include:

1. **SIFT:** Creates distinctive descriptors that are robust to scale and rotation changes.
2. **SURF:** Uses local image information to generate fast descriptors.
3. **BRIEF:** Creates binary descriptors by comparing pixel intensities.
4. **ORB:** Combines oriented FAST keypoints with BRIEF descriptors.

3. Feature Matching Process

The process involves the following steps:

Step 1: Image Acquisition

Take two images containing the same object or scene from different viewpoints.

Step 2: Feature Detection

Detect important keypoints using algorithms such as SIFT, SURF or ORB.

Step 3: Descriptor Extraction

Generate numerical descriptors for every detected keypoint.

Step 4: Descriptor Matching

Compare descriptors from both images using distance measurements.

Common methods: - Euclidean Distance. - Hamming Distance.

Step 5: Match Selection

Select the best matching pairs using techniques such as the Nearest Neighbour and Ratio Test.

Step 6: Outlier Removal

Use RANSAC (Random Sample Consensus) to remove incorrect matches.

Step 7: Final Matching

Display the correct matching points between the two images.

4. Distance Formula

For two feature descriptors, Euclidean Distance is:

[D(A,B)=\sqrt{\sum_{i=1}^{n}(A_i-B_i)^2}]

Where: - (A,B) = Two feature descriptors. - (n) = Number of descriptor elements. - (D) = Distance between descriptors.

A smaller distance indicates greater similarity.

5. Applications

1. Image stitching and panorama creation.
2. Object recognition.
3. 3D reconstruction.
4. Motion tracking.
5. Image registration.

6. Advantages

- Identifies similar features across images.
- Works with changes in scale and rotation when robust descriptors are used.
- Helps improve object recognition accuracy.

Conclusion: Feature matching uses feature descriptors to identify corresponding points between images. It is an important technique in Computer Vision for image alignment, object recognition and 3D reconstruction.

Unit 2 • Q4

Compare Point, Line and Edge Features with Suitable Examples.

> 📝 **Exam Question:** 4. Compare point, line, and edge features with suitable examples.

Unit II – Q4. Compare Point, Line and Edge Features with Suitable Examples.

1. Introduction

In Computer Vision, **Point, Line and Edge Features** are important image features used to identify objects, shapes and boundaries. They help computers understand the structure and important details of an image.

2. Types of Features

1. Point Features

Point features represent specific locations in an image that contain important visual information.

- Represented using coordinates (x, y).
- Usually identify corners or distinctive keypoints.
- Used in image matching and object tracking.

Examples: Harris Corner Detector, SIFT and FAST.

2. Line Features

Line features represent straight or nearly straight structures in an image.

- Connect multiple points.
- Represent object boundaries and structural patterns.
- Used in road detection and shape analysis.

Examples: Hough Transform and Line Segment Detector (LSD).

3. Edge Features

Edge features represent boundaries where image intensity or colour changes suddenly.

- Identify object boundaries.
- Help separate objects from backgrounds.
- Used in image segmentation and shape detection.

Examples: Sobel Operator, Canny Edge Detector and Prewitt Operator.

3. Diagram

4. Comparison Between Point, Line and Edge Features

Parameter	Point Features	Line Features	Edge Features
Meaning	Specific image locations	Straight structures	Intensity boundaries
Representation	Coordinates	Connected points	Boundary pixels
Main purpose	Identify keypoints	Detect lines	Detect object boundaries

Parameter	Point Features	Line Features	Edge Features
Example	Harris Detector	Hough Transform	Canny Detector
Application	Object matching	Road detection	Image segmentation

5. Suitable Examples

1. **Point:** Detecting the four corners of a window in a building image.
2. **Line:** Detecting road lane markings in a traffic image.
3. **Edge:** Identifying the boundary between a person and the background.

6. Applications

- Object recognition.
- Image segmentation.
- Image matching.
- Motion tracking.
- 3D reconstruction.
- Road and shape detection.

Conclusion: Point features identify important locations, line features represent straight structures, and edge features identify object boundaries. All three are essential for feature extraction and image analysis in Computer Vision.

Unit 2 • Q5

Explain the Steps Involved in the Canny Edge Detector Algorithm.

> 📝 **Exam Question:** 5. Explain the steps involved in the Canny Edge Detector algorithm



FIGURE 2.2: Canny Edge Detection Algorithm 5-Step Pipeline

Systematic pipeline guaranteeing good detection, low error rate, and single-pixel response.

Unit II – Q5. Explain the Steps Involved in the Canny Edge Detector Algorithm.

1. Definition of Canny Edge Detector

Canny Edge Detector is an edge detection algorithm used in Computer Vision to identify object boundaries in an image. It was developed by John F. Canny in 1986.

It detects edges accurately while reducing noise and unwanted responses.

2. Steps Involved in Canny Edge Detection

Step 1: Noise Reduction (Gaussian Filtering)

- Apply a Gaussian filter to the input image.
- Removes unwanted noise.
- Prevents incorrect edge detection.

Step 2: Gradient Calculation

- Calculate image intensity changes in horizontal and vertical directions.
- Uses Sobel operators.
- Identifies edge strength and direction.

Gradient magnitude:

[G=\sqrt{G_x^2+G_y^2}]

Gradient direction:

[\theta=\tan^{-1}\left(\frac{G_y}{G_x}\right)]

Where (G_x) and (G_y) represent horizontal and vertical gradients.

Step 3: Non-Maximum Suppression

- Removes unnecessary edge pixels.
- Keeps only pixels having maximum gradient values.
- Produces thin and sharp edges.

Step 4: Double Thresholding

- Uses two threshold values: High and Low.
- Classifies detected edges into three categories:
- Strong edges.
- Weak edges.
- Non-edges.

Step 5: Edge Tracking by Hysteresis

- Connects weak edges to strong edges if they are connected.
- Removes weak edges that are not connected to strong edges.
- Produces the final edge image.

3. Working Flowchart

Input Image Gaussian Filtering Gradient Calculation Non-Maximum Suppression Double Thresholding Edge Tracking by Hysteresis Final Edge Image

4. Applications

1. Object detection and recognition.
2. Medical image processing.
3. Road and lane detection.
4. Image segmentation.
5. Shape identification.

5. Advantages

- Produces thin and accurate edges.
- Reduces noise.
- Minimizes false edge detection.
- Provides good edge localization.

Conclusion: Canny Edge Detector uses five major steps: Gaussian filtering, gradient calculation, non-maximum suppression, double thresholding and hysteresis. It is widely used for accurate edge detection in Computer Vision.

Unit 2 • Q6

Describe Split and Merge Segmentation Techniques with Suitable Examples.

> 📝 **Exam Question:** 6. Describe split and merge segmentation techniques with suitable examples.

Unit II – Q6. Describe Split and Merge Segmentation Techniques with Suitable Examples.

1. Definition of Split and Merge Segmentation

Split and Merge Segmentation is a region-based image segmentation technique that divides an image into smaller regions and combines similar regions based on their properties, such as colour, intensity and texture.

It uses a **Homogeneity Criterion** to decide whether a region should be split or merged.

2. Working of Split and Merge Technique

This technique involves four main steps:

Step 1: Initial Image

- Take the complete image as one region.
- Check whether all pixels satisfy a similarity condition.

Step 2: Splitting

- If a region is not homogeneous, divide it into four smaller regions.
- This process continues until each region satisfies the required condition or reaches a minimum size.

Step 3: Merging

- Check neighbouring regions for similarity.
- Merge regions if they satisfy the homogeneity condition.

Step 4: Final Segmentation

- Produce the final image containing meaningful regions.
- Each region represents similar image properties.

3. Diagram of Split and Merge Technique

4. Mathematical Representation

A region (R) is homogeneous if it satisfies:

[P(R)=TRUE]

Where (P(R)) is a similarity condition based on pixel properties.

For example:

[|I_{\max}-I_{\min}|<T]

Where: - (I_{\max}) = Maximum intensity. - (I_{\min}) = Minimum intensity. - (T) = Threshold value.

If the condition is false, the region is split. If neighbouring regions satisfy the similarity condition, they are merged.

5. Suitable Example

Consider an image containing a red ball on a blue background.

1. Initially, the complete image is considered one region.
2. Since the image contains different colours, it is divided into four regions.
3. Regions containing similar colours are further processed.
4. Similar neighbouring regions are merged.
5. Finally, the red ball and blue background are separated into meaningful regions.

6. Applications

1. Medical image segmentation.
2. Satellite image analysis.
3. Object detection.
4. Image compression.
5. Agricultural image processing.

7. Advantages and Limitations

Advantages: - Simple and easy to implement. - Useful for detecting homogeneous regions. - Does not require edge detection.

Limitations: - Can produce block-shaped boundaries. - Results depend on threshold selection. - May require repeated splitting and merging.

Conclusion: Split and Merge is a region-based segmentation technique that divides non-uniform regions and combines similar neighbouring regions to produce meaningful image segments.

Unit 2 • Q7

Explain the Mean Shift Algorithm and Mode Finding Technique Used in Segmentation.

> 📝 **Exam Question:** 7. Explain the mean shift algorithm and mode finding technique used in segmentation.

FORMULA **Mean Shift Gradient Vector Formula**

$$\mathbf{m}(\mathbf{x}) = \frac{\sum_{i=1}^{n} K(\mathbf{x} - \mathbf{x}_i)\mathbf{x}_i}{\sum_{i=1}^{n} K(\mathbf{x} - \mathbf{x}_i)} - \mathbf{x}$$

Directs the shifting window along the local density gradient toward the nearest mode.

Unit II – Q7. Explain the Mean Shift Algorithm and Mode Finding Technique Used in Segmentation.

1. Definition of Mean Shift Algorithm

Mean Shift is a non-parametric clustering algorithm used in Computer Vision for image segmentation and object tracking. It identifies regions containing similar pixels by finding high-density areas in the feature space.

It groups pixels based on similarities in colour, intensity and spatial location without requiring a predefined number of clusters.

2. Mode Finding Technique

Mode Finding is the process of identifying the points where data density is maximum.

In Mean Shift, each data point moves towards the nearest high-density region called a **Mode**.

For example, pixels having similar colours move towards the same density centre and form a cluster.

3. Working Steps of Mean Shift Algorithm

Step 1: Feature Space Representation

Represent image pixels using features such as colour and spatial coordinates.

Step 2: Select a Window

Choose a window called the **Kernel Window** around a data point.

Step 3: Calculate Mean

Calculate the mean position of all points inside the window.

Step 4: Shift the Window

Move the window towards the calculated mean position, where data density is higher.

Step 5: Repeat the Process

Repeat the calculation and shifting until the window position becomes stable.

Step 6: Identify Modes

Points converging to the same mode are grouped into one cluster.

Step 7: Image Segmentation

Assign similar pixels to the same region to produce the segmented image.

4. Mean Shift Mathematical Formula

The Mean Shift vector is calculated as:

[M(x)= \frac{\sum_{i=1}^{n}K(x_i-x)x_i} {\sum_{i=1}^{n}K(x_i-x)}-x]

Where:

- (x) = Current position.
- (x_i) = Data points.
- (K) = Kernel function.
- (M(x)) = Mean Shift vector.

The window moves in the direction of the Mean Shift vector until it reaches a density peak.

5. Example

Consider an image containing red, green and blue objects.

1. Select pixels based on their colour and position.
2. Calculate the mean of neighbouring pixels.
3. Move pixels towards their nearest density peaks.
4. Group similar colours into clusters.
5. Obtain separate red, green and blue regions.

6. Applications

1. Image segmentation.
2. Object tracking.
3. Colour-based clustering.
4. Object detection.
5. Image analysis.

7. Advantages

- Does not require a predefined number of clusters.
- Can detect clusters of different shapes.
- Useful for image segmentation and tracking.

Conclusion: Mean Shift is a density-based clustering technique that finds modes by repeatedly shifting data points towards high-density regions. It is useful for grouping similar pixels and segmenting images.

Unit 2 • Q8

Describe Feature Matching and Feature Tracking Techniques and Explain Their Applications in Real-World Vision Systems.

> 📝 **Exam Question:** 8. Describe feature matching and feature tracking techniques and explain their applications in real-world vision systems.

Unit II – Q8. Describe Feature Matching and Feature Tracking Techniques and Explain Their Applications in Real-World Vision Systems.

1. Feature Matching

Feature Matching is the process of identifying and matching similar features between two or more images. It uses keypoints and feature descriptors to find corresponding points in different images.

It helps computers recognize the same object even when the images have different scales, rotations or viewpoints.

Techniques of Feature Matching:

1. **SIFT:** Detects distinctive keypoints and generates descriptors that are robust to scale and rotation changes.
2. **SURF:** Uses Hessian Matrix and Integral Images for fast feature detection and matching.
3. **ORB:** Combines FAST keypoint detection with BRIEF descriptors for efficient matching.
4. **Brute Force Matching:** Compares each descriptor in one image with descriptors in another image to find the closest match.
5. **FLANN:** Uses fast approximate nearest-neighbour searching for matching large numbers of descriptors.

2. Feature Tracking

Feature Tracking is the process of continuously identifying and following important points or features across multiple video frames.

It helps determine the movement, position and direction of objects over time.

Techniques of Feature Tracking:

1. **Lucas-Kanade Method:** Tracks feature points by estimating their movement between consecutive frames using optical flow.
2. **Optical Flow:** Estimates the apparent movement of pixels between video frames.
3. **Mean Shift:** Tracks objects by moving towards regions of higher feature density.
4. **Kalman Filter:** Predicts and updates object positions using previous movement information.

3. Working Process

Feature Matching Steps: 1. Capture two images. 2. Detect important keypoints. 3. Extract feature descriptors. 4. Compare descriptors. 5. Select matching points.

Feature Tracking Steps: 1. Detect features in the first video frame. 2. Identify their positions. 3. Track the same features in subsequent frames. 4. Calculate their movement. 5. Update their positions continuously.

4. Difference Between Feature Matching and Tracking

Feature Matching	Feature Tracking
Compares different images	Follows features across frames
Finds corresponding points	Estimates continuous movement
Used in image stitching	Used in video analysis
Example: SIFT	Example: Lucas-Kanade

5. Real-World Applications

1. **Autonomous Vehicles:** Tracks pedestrians, vehicles and road features.
2. **Robotics:** Helps robots understand movement and surroundings.
3. **Augmented Reality:** Maintains virtual objects at correct positions.
4. **Video Surveillance:** Tracks moving people and objects.
5. **Panorama Creation:** Matches overlapping images to create panoramic views.
6. **3D Reconstruction:** Matches features across multiple images to estimate 3D structures.

Conclusion: Feature matching identifies corresponding points between images, whereas feature tracking follows those points over time. Both techniques are essential for object recognition, motion estimation, robotics and autonomous vision systems.

Unit 2 • Q9

Explain Feature Detectors and Feature Descriptors and Compare Their Roles in Computer Vision.

> **Exam Question:** 9. Explain feature detectors and feature descriptors and compare their roles in computer vision.

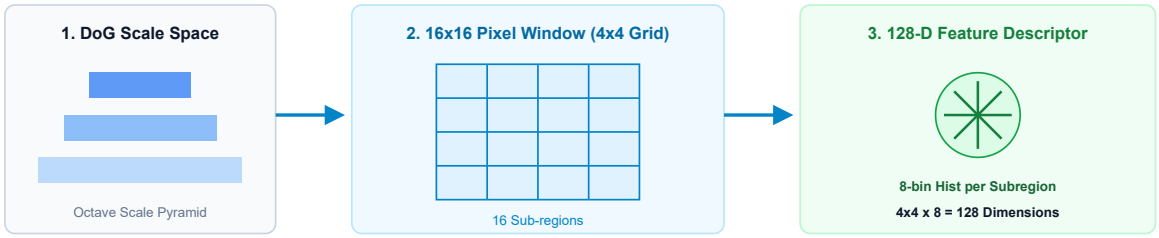


FIGURE 2.3: SIFT Scale-Space Detection and 128-D Descriptor Generation

Difference of Gaussians detects extrema; local 16x16 gradient neighborhood compiles into 128-dimensional unit vector.

Unit II – Q9. Explain Feature Detectors and Feature Descriptors and Compare Their Roles in Computer Vision.

1. Feature Detectors

Feature Detectors are algorithms used to identify important points or regions in an image, such as corners, edges and keypoints.

They locate distinctive areas that can help computers recognize and analyze objects.

Types of Feature Detectors:

1. **Harris Corner Detector:** Detects corners where image intensity changes in two directions.
2. **SIFT:** Detects keypoints at different image scales using Difference of Gaussian (DoG).
3. **SURF:** Uses Hessian Matrix to detect important image features.
4. **FAST:** Detects corners quickly by comparing surrounding pixel intensities.
5. **ORB:** Uses FAST for keypoint detection and assigns orientation information.

2. Feature Descriptors

Feature Descriptors are numerical representations of detected keypoints. They describe the local appearance around a feature so that it can be compared with features in other images.

Types of Feature Descriptors:

1. **SIFT Descriptor:** Generates a 128-dimensional feature vector using local gradient information.
2. **SURF Descriptor:** Uses Haar wavelet responses to describe local image features.
3. **BRIEF:** Creates binary descriptors using comparisons between pairs of pixel intensities.
4. **ORB Descriptor:** Uses an oriented version of BRIEF to provide rotation robustness.

3. Working Process

The process involves:

1. **Input Image:** Capture an image.
2. **Feature Detection:** Identify important keypoints.
3. **Descriptor Extraction:** Generate numerical descriptions of detected keypoints.
4. **Feature Matching:** Compare descriptors between images.
5. **Recognition:** Identify matching objects or image regions.

4. Comparison Between Feature Detectors and Descriptors

Feature Detectors	Feature Descriptors
Locate important image points	Describe detected image points
Identify corners and keypoints	Generate numerical representations
Answer where features exist	Answer what features look like
Example: Harris, FAST	Example: BRIEF, SIFT descriptor
Used for feature localization	Used for feature comparison

5. Importance in Computer Vision

1. **Object Recognition:** Helps identify objects in different images.
2. **Image Stitching:** Finds matching points between overlapping images.
3. **Object Tracking:** Helps follow objects across video frames.
4. **3D Reconstruction:** Finds corresponding points from multiple viewpoints.
5. **Image Registration:** Aligns two or more images correctly.

6. Example

Suppose two photographs contain the same building from different viewpoints.

- The **Feature Detector** identifies important corners and keypoints in both images.
- The **Feature Descriptor** creates numerical descriptions of those keypoints.
- The matching algorithm compares the descriptors and identifies corresponding points.

Conclusion: Feature detectors locate important points in an image, while feature descriptors describe those points numerically. Both work together to perform accurate feature matching, object recognition and image analysis.

QUICK REVISION / COMPLETE RECAP

Unit 2: Image Segmentation & Feature Extraction — Complete Recap

High-yield revision summary, definitions, key points to remember, and core formulas for last-minute exam prep.

1. Unit Overview & Summary

Unit 2 covers mid-level computer vision: segmenting images into meaningful semantic regions, detecting distinctive keypoints and edges, generating invariant feature descriptors, and robustly matching and tracking features across frames.

2. Core Concepts & Definitions

- **Image Segmentation**: Partitioning an image into non-overlapping, homogeneous regions that share uniform visual characteristics (intensity, color, texture).
- **Graph Cut Segmentation**: Formulates segmentation as an s-t graph cut problem on a directed graph $G = (V, E)$. Pixels are internal nodes, and two terminal nodes represent source (s, foreground) and sink (t, background). The min-cut max-flow theorem (Boykov-Kolmogorov) finds the global minimum cut in polynomial time, minimizing energy:

  FORMULA Governing Mathematical Equation

  $$E(L) = \sum_{p \in V} D_p(L_p) + \sum_{(p,q) \in \mathcal{N}} V_{p,q}(L_p, L_q)$$

- **Normalized Cuts (N-cut)**: Normalized cut (Shi & Malik) overcomes the standard min-cut bias toward isolating tiny clusters of outliers by normalizing the cut cost by total association with the whole graph:

  FORMULA Governing Mathematical Equation

  $$\text{Ncut}(A, B) = \frac{\text{cut}(A, B)}{\text{assoc}(A, V)} + \frac{\text{cut}(A, B)}{\text{assoc}(B, V)}$$

Solved as a relaxed continuous generalized eigensystem: $(D - W)y = \lambda D y$. - **Mean Shift Algorithm**: A non-parametric iterative mode-seeking algorithm based on Kernel Density Estimation (KDE). Computes the mean shift vector $\mathbf{m}(\mathbf{x})$ pointing in the direction of maximum increase in local density:

FORMULA Governing Mathematical Equation

$$\mathbf{m}(\mathbf{x}) = \frac{\sum_{i=1}^{n} K(\mathbf{x} - \mathbf{x}_i)\mathbf{x}_i}{\sum_{i=1}^{n} K(\mathbf{x} - \mathbf{x}_i)} - \mathbf{x}$$

Points converging to the same local density mode are assigned to the same cluster. - **Split and Merge Segmentation**: A region-based top-down and bottom-up quadtree segmentation algorithm: - *Split*: Recursively subdivide non-homogeneous quadrants until all leaf nodes satisfy homogeneity condition $P(R_i) = \text{TRUE}$. - *Merge*: Merge adjacent regions R_j, R_k if joint region satisfies $P(R_j \cup R_k) = \text{TRUE}$. - **Harris Corner Detector**: Detects

interest points by measuring intensity variance in a local window under shifts $(\Delta x, \Delta y)$ via the second-moment structure matrix $\mathbf{M}$:

FORMULA **Governing Mathematical Equation**

$$\mathbf{M} = \sum_{(x,y) \in W} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \quad R = \det(\mathbf{M}) - k(\text{trace}(\mathbf{M}))^2$$

- $R > 0$: Corner point.
- $R < 0$: Edge region.
- $|R| \approx 0$: Flat region.
- **SIFT (Scale-Invariant Feature Transform)**:
- *Scale-Space Extrema Detection*: Detects scale-invariant keypoints using Difference of Gaussians (DoG): $D(x, y, \sigma) = (G(x,y,k\sigma) - G(x,y,\sigma)) * I(x,y)$.
- *Keypoint Localization*: Sub-pixel Taylor expansion and elimination of low-contrast points and unstable edge responses.
- *Orientation Assignment*: Assigns dominant gradient orientation to achieve rotation invariance.
- *Descriptor Generation*: 4×4 spatial grid of 8-bin gradient histograms $\implies 128$-dimensional descriptor vector normalized to unit length.
- **SURF (Speeded-Up Robust Features)**: Accelerates SIFT by approximating Laplacian of Gaussian using box filters and integral images, producing a compact 64-D descriptor.
- **Feature Matching Strategy**:
- *Nearest Neighbor Distance Ratio (NNDR)*: Rejects ambiguous matches if $\frac{d_1}{d_2} > 0.8$ (where d_1 is distance to 1st nearest neighbor, d_2 is distance to 2nd nearest neighbor).
- *Cross-checking / Bidirectional validation*: Mutual nearest neighbors only.
- **Feature Tracking**:
- *Kanade-Lucas-Tomasi (KLT) Tracker*: Tracks Shi-Tomasi good features across video frames using local Taylor expansion of optical flow.

---

3. Canny Edge Detection Algorithm (5 Steps)

1. **Gaussian Smoothing**: Filter input image with Gaussian kernel to suppress high-frequency noise.
2. **Gradient Computation**: Compute gradient magnitudes $M(x,y) = \sqrt{G_x^2 + G_y^2}$ and directions $\theta = \arctan(G_y / G_x)$ using Sobel filters.
3. **Non-Maximum Suppression (NMS)**: Thin edges down to 1-pixel width by retaining only pixels that are local maxima along the gradient direction.
4. **Double Thresholding**: Classify pixels into strong edges ($> T_{\text{high}}$), weak edges ($T_{\text{low}} \le \cdot \le T_{\text{high}}$), and suppressed non-edges ($< T_{\text{low}}$).
5. **Edge Tracking by Hysteresis**: Retain weak edge pixels only if they are 8-connected to a strong edge pixel.

---

4. Key Differences Summary Table

Feature	Harris Corner	SIFT	SURF
Scale Invariance	No (Fixed scale)	Yes (DoG scale space)	Yes (Fast Hessian / Box filter)
Rotation Invariance	Yes	Yes (Dominant gradient)	Yes (Haar wavelets)
Descriptor Size	None (detector only)	128 dimensions	64 dimensions

Feature	Harris Corner	SIFT	SURF
Speed	Very Fast	Moderate	Fast

✎ SHORT ANSWER QUESTIONS (2 MARKS)

Unit 2 — 2-Mark Exam Questions

Crisp, direct, point-to-point answers designed for maximum marks in university short-answer sections.

Q1 What is Image Segmentation?

Image Segmentation is the process of dividing an image into multiple meaningful regions or segments based on similarities in colour, intensity or texture. It helps identify objects and boundaries. It is used in medical imaging, object detection and autonomous vehicles.

Q2 What is Graph Cut?

Graph Cut is an image segmentation technique used to separate foreground objects from the background. It represents an image as a graph containing nodes and edges. By finding the minimum cut, it separates pixels into different regions based on similarity and boundary information.

Q3 What is Normalized Cuts?

Normalized Cuts is a graph-based image segmentation technique that divides an image into meaningful regions. It represents pixels as graph nodes and similarities as weighted edges. It minimizes connections between different groups while maintaining strong connections within the same group.

Q4 What is Mean Shift?

Mean Shift is a non-parametric clustering algorithm used in computer vision to identify groups of similar data points. It repeatedly moves a window towards the region of highest data density. It is commonly used in object tracking, image segmentation and feature detection.

Q5 What is SIFT (Scale-Invariant Feature Transform) and why is it used?

SIFT (Scale-Invariant Feature Transform) is an algorithm developed by David Lowe that detects distinctive local keypoints and computes 128-dimensional gradient orientation descriptors using Difference of Gaussians (DoG). It is widely used because it remains robustly invariant to image scale, 2D rotation, illumination changes, and affine distortions, making it ideal for object recognition, feature matching, and panorama stitching.

Q6 Briefly Explain SURF Method.

SURF (Speeded-Up Robust Features) is a feature detection and description algorithm used in computer vision. It uses **Hessian Matrix** and **Integral Images** to detect keypoints quickly. It provides scale and rotation invariance and is commonly used for image matching and object recognition.

Q7 **What is Object Tracking?**

Object Tracking is the process of detecting and continuously following an object's movement across multiple video frames. It identifies the object's position, direction and speed over time. It is used in video surveillance, traffic monitoring, sports analysis and autonomous vehicles.

UNIT 3

Unit 3: Motion Estimation & Structure from Motion (SfM)

Comprehensive Question & Answer Collection • Quick Revision Recap • 2-Mark Direct Questions

PART A: DETAILED EXAMINATION QUESTIONS & ANSWERS

Unit 3 • Q1

Explain Two-Frame and Factorization Structure from Motion with Neat Diagram.

> **Exam Question:** **Unit-III** 1\. Explain Two-Frame and factorization Structure from Motion with neat diagram.

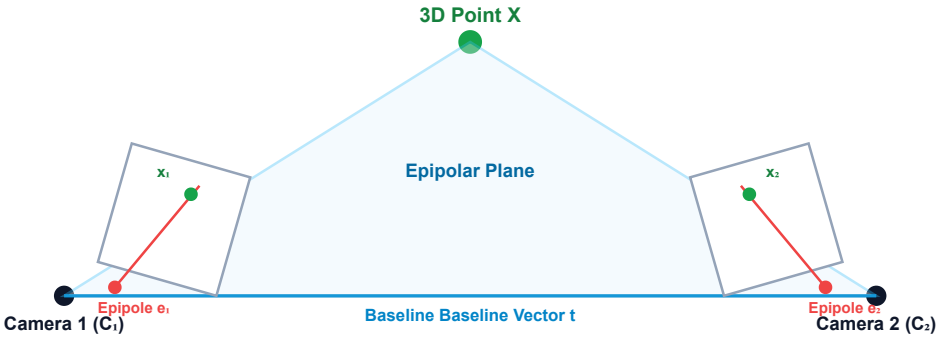


FIGURE 3.1: Two-Frame Epipolar Geometry Architecture

Constraint $x_2^T \cdot F \cdot x_1 = 0$ restricts correspondence search from a 2D image plane to a 1D epipolar line.

FORMULA Essential Matrix Epipolar Constraint (5 DOF)

$$\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}, \quad \mathbf{x}^T \mathbf{E} \mathbf{x} = 0$$

Encapsulates calibrated relative rotation R and translation t; non-zero singular values are equal.

FORMULA Fundamental Matrix Epipolar Constraint (7 DOF)

$$\mathbf{F} = \mathbf{K}^{-T} \mathbf{E} \mathbf{K}^{-1}, \quad \mathbf{x}^T \mathbf{F} \mathbf{x} = 0$$

Applies to uncalibrated camera views; satisfies det(F) = 0 and is estimated via the 8-Point Algorithm.

Unit III – Q1. Explain Two-Frame and Factorization Structure from Motion with Neat Diagram.

1. Structure from Motion (SfM)

Structure from Motion (SfM) is a computer vision technique used to reconstruct the 3D structure of a scene from multiple 2D images taken from different camera positions.

It estimates both the **3D positions of objects (Structure)** and **camera movement (Motion)**.

There are two important approaches: Two-Frame SfM and Factorization SfM.

2. Two-Frame Structure from Motion

Two-Frame Structure from Motion is a technique that estimates the 3D structure of a scene using two images captured from different camera positions.

Working Steps:

1. **Image Acquisition:** Capture two images of the same scene from different viewpoints.
2. **Feature Detection:** Detect important keypoints using SIFT or ORB.
3. **Feature Matching:** Find corresponding feature points in both images.
4. **Camera Motion Estimation:** Estimate camera rotation and translation using the Essential Matrix.
5. **Triangulation:** Calculate 3D point positions using corresponding image points.
6. **3D Reconstruction:** Generate a 3D representation of the scene.

Diagram:

Example: Taking two photographs of a building from different positions and reconstructing its 3D structure.

3. Factorization Structure from Motion

Factorization Structure from Motion is a technique used to recover 3D structure and camera motion from multiple images by decomposing a measurement matrix.

It was introduced by Tomasi and Kanade.

Working Steps:

1. **Image Acquisition:** Capture multiple images of a moving camera observing a static object.
2. **Feature Tracking:** Identify and track common feature points across all images.
3. **Measurement Matrix:** Create a matrix containing the 2D coordinates of tracked points.
4. **Matrix Centering:** Subtract the mean coordinates to remove translation effects.
5. **Matrix Factorization:** Apply Singular Value Decomposition (SVD) to decompose the measurement matrix.
6. **3D Reconstruction:** Estimate camera motion and 3D point coordinates.

Mathematical Representation:

[W=MS]

Where:

- (W) = Measurement matrix.
- (M) = Camera motion matrix.
- (S) = 3D structure matrix.

Using SVD:

[W=U\Sigma V^T]

Diagram:

4. Comparison

Two-Frame SfM	Factorization SfM
Uses two images	Uses multiple images
Estimates relative camera motion	Estimates motion across multiple views
Uses Essential Matrix and triangulation	Uses measurement matrix and SVD
Suitable for two-view reconstruction	Suitable for multi-view reconstruction

5. Applications

1. 3D object reconstruction.
2. Robotics and navigation.
3. Augmented Reality.
4. 3D mapping.
5. Virtual reality.

Conclusion: Two-Frame SfM reconstructs 3D structure using two images, whereas Factorization SfM uses multiple images and matrix decomposition to estimate camera motion and 3D structure.

Unit 3 • Q2

Analyze Triangulation and Bundle Adjustment Methods in Structure from Motion.

> **Exam Question:** 2. Analyze Triangular and bundle adjustment methods in structure from motion.

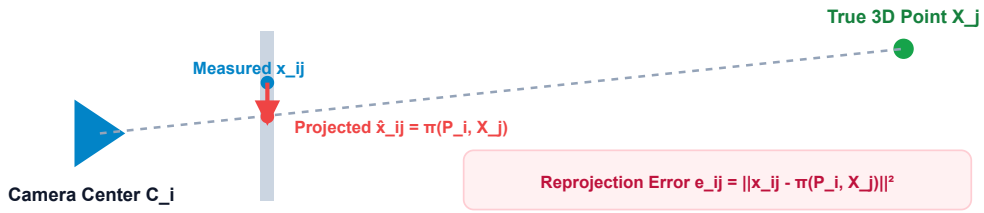


FIGURE 3.2: Triangulation and Reprojection Error in Bundle Adjustment

Bundle adjustment minimizes the cumulative Euclidean distance between observed 2D feature coordinates and reprojected 3D points.

FORMULA **Bundle Adjustment Non-Linear Reprojection Objective**

$$\min_{\{\mathbf{P}_i\},\{\mathbf{X}_j\}} \sum_{i=1}^{m} \sum_{j=1}^{n} v_{ij} \left\| \mathbf{x}_{ij} - \pi(\mathbf{P}_i, \mathbf{X}_j) \right\|^2$$

Jointly optimizes camera matrices P_i and 3D points X_j via Levenberg-Marquardt optimization.

Unit III – Q2. Analyze Triangulation and Bundle Adjustment Methods in Structure from Motion.

1. Introduction to Structure from Motion (SfM)

Structure from Motion (SfM) is a computer vision technique used to reconstruct the 3D structure of a scene from multiple 2D images captured from different camera positions.

Two important methods used in SfM are **Triangulation** and **Bundle Adjustment**.

2. Triangulation Method

Triangulation is the process of calculating the 3D position of a point using its corresponding 2D positions in two or more images taken from different camera viewpoints.

It uses camera positions, orientations and matched feature points to estimate depth and 3D coordinates.

Working Steps:

1. **Image Acquisition:** Capture two or more images from different viewpoints.
2. **Feature Detection:** Identify important keypoints using SIFT or ORB.
3. **Feature Matching:** Find corresponding points between images.
4. **Camera Estimation:** Estimate camera positions and orientations.
5. **Ray Intersection:** Project rays from camera centres through corresponding image points.
6. **3D Point Calculation:** Estimate the 3D position where the rays intersect or come closest.

Diagram:

Mathematical Representation:

[P_1 \leftrightarrow P_2 \rightarrow P(X,Y,Z)]

Where: - (P_1,P_2) = Corresponding image points. - (P(X,Y,Z)) = Reconstructed 3D point.

Example: Reconstructing the 3D position of a building corner using two photographs taken from different positions.

3. Bundle Adjustment Method

Bundle Adjustment (BA) is an optimization technique used in Structure from Motion to improve the accuracy of 3D reconstruction.

It simultaneously refines camera positions, camera orientations and 3D point coordinates by minimizing reprojection error.

Working Steps:

1. **Initial Reconstruction:** Obtain initial camera parameters and 3D points.
2. **Projection:** Project 3D points onto the image planes.
3. **Error Calculation:** Calculate the difference between observed and projected image points.
4. **Optimization:** Adjust camera parameters and 3D coordinates.
5. **Error Minimization:** Repeat the process until reprojection error is minimized.
6. **Final Reconstruction:** Obtain an improved 3D model.

Diagram:

Reprojection Error Formula:

[E=\sum_{i,j}\left|x_{ij}-\hat{x}_{ij}\right|^2]

Where:

- (x_{ij}) = Observed image point.
- (\hat{x}_{ij}) = Projected image point.
- (E) = Total reprojection error.

4. Comparison

Triangulation	Bundle Adjustment
Calculates 3D points	Refines 3D reconstruction
Uses multiple camera views	Uses multiple cameras and points
Estimates depth	Minimizes reprojection error
Produces initial 3D structure	Improves reconstruction accuracy

5. Applications

1. 3D object reconstruction.
2. Robotics and navigation.
3. Drone-based mapping.
4. Augmented Reality.
5. 3D scene modelling.

Conclusion: Triangulation estimates 3D point positions from multiple images, while Bundle Adjustment improves the reconstruction by minimizing reprojection error. Both methods are essential for accurate Structure from Motion systems.

Unit 3 • Q3

Compare Two-Frame and Constrained Structure from Motion.

> 📝 **Exam Question:** 3. Compare Two Frame and Constrained structure from motions.

Unit III – Q3. Compare Two-Frame and Constrained Structure from Motion.

1. Two-Frame Structure from Motion

Two-Frame Structure from Motion (SfM) is a technique used to estimate the 3D structure of a scene and relative camera motion using two images captured from different viewpoints.

Working Steps:

1. Capture two images of the same scene.
2. Detect keypoints using SIFT or ORB.
3. Match corresponding features between images.
4. Estimate camera motion using the Essential Matrix.
5. Apply triangulation to calculate 3D point positions.
6. Generate the 3D reconstruction.

Example: Reconstructing a building using two photographs taken from different camera positions.

2. Constrained Structure from Motion

Constrained Structure from Motion is a 3D reconstruction technique that uses additional information or restrictions to improve the accuracy and stability of camera motion and 3D structure estimation.

Constraints provide extra information about the scene or camera movement.

Types of Constraints:

1. **Geometric Constraints:** Use known geometric relationships between objects.
2. **Motion Constraints:** Restrict camera movement or orientation.
3. **Camera Constraints:** Use known camera parameters.
4. **Scene Constraints:** Use information such as parallel lines or known distances.

Working Steps:

1. Capture multiple images of a scene.
2. Detect and match common features.
3. Estimate initial camera positions and 3D points.
4. Apply known geometric or camera constraints.
5. Optimize camera motion and 3D structure.
6. Generate an improved 3D model.

Example: Reconstructing a building using images with known camera calibration and parallel architectural lines.

3. Comparison Between Two-Frame and Constrained SfM

Parameter	Two-Frame SfM	Constrained SfM
Images	Uses two images	Can use multiple images

Parameter	Two-Frame SfM	Constrained SfM
Main purpose	Estimates 3D structure and relative motion	Improves estimation using constraints
Information	Image feature correspondences	Correspondences plus additional constraints
Accuracy	Depends on image quality and viewpoint	Can improve accuracy with reliable constraints
Complexity	Relatively simple	Depends on constraints used
Applications	Basic 3D reconstruction	Accurate scene reconstruction

4. Applications

1. 3D object reconstruction.
2. Robotics and navigation.
3. Architectural modelling.
4. Autonomous vehicles.
5. Augmented Reality.
6. 3D mapping.

5. Advantages

- **Two-Frame SfM:** Simple and useful for estimating relative camera motion and initial 3D structure.
- **Constrained SfM:** Additional information can reduce ambiguity and improve reconstruction accuracy.

Conclusion: Two-Frame SfM estimates 3D structure using two images, whereas Constrained SfM uses additional geometric, camera or motion information to improve reconstruction accuracy and reliability.

Unit 3 • Q4

Apply Translation Alignment and Parametric Motion in Dense Motion Estimation with an Example.

> 📝 **Exam Question:** 4. Apply translation alignment and parametric motion in dense motion estimation with an example.

Unit III – Q4. Apply Translation Alignment and Parametric Motion in Dense Motion Estimation with an Example.

1. Dense Motion Estimation

Dense Motion Estimation is a computer vision technique used to calculate the movement of every pixel between two consecutive video frames. It helps understand how objects and camera positions change over time.

It is commonly used in video analysis, object tracking and motion detection.

2. Translation Alignment

Translation Alignment is a technique used to align two images by shifting one image horizontally or vertically without changing its shape, size or orientation.

It estimates the movement using two parameters:

- (t_x) = Horizontal translation.
- (t_y) = Vertical translation.

Mathematical Formula:

[x'=x+t_x]

[y'=y+t_y]

Working Steps:

1. Take two consecutive images.
2. Identify corresponding pixels or features.
3. Calculate horizontal and vertical displacement.
4. Shift the second image to align with the first.
5. Estimate the remaining pixel movement.

Example: If a car moves 10 pixels to the right and 5 pixels downward, its translation is:

[t_x=10,\quad t_y=5]

3. Parametric Motion

Parametric Motion represents image movement using a mathematical model with a limited number of parameters.

It describes the overall motion of objects or the camera using transformations such as translation, rotation, scaling and affine motion.

Affine Motion Formula:

[\begin{bmatrix} x'\y' \end{bmatrix} = \begin{bmatrix} a&b\c&d \end{bmatrix} \begin{bmatrix} x\y \end{bmatrix} + \begin{bmatrix} t_x\t_y \end{bmatrix}]

Where:

- (a,b,c,d) = Motion transformation parameters.
- (t_x,t_y) = Translation parameters.
- (x,y) = Original pixel coordinates.
- (x',y') = New pixel coordinates.

Working Steps:

1. Capture two consecutive frames.
2. Detect corresponding feature points.
3. Estimate motion parameters.
4. Apply the transformation model.
5. Calculate pixel displacement.
6. Generate the motion estimation result.

4. Difference Between Translation and Parametric Motion

Translation Alignment	Parametric Motion
Uses horizontal and vertical shifts	Uses multiple motion parameters
Simple motion model	More flexible motion model
Does not change orientation or size	Can represent rotation and scaling
Uses two parameters	Affine model uses six parameters

5. Applications

1. Video stabilization.
2. Object tracking.
3. Motion detection.
4. Autonomous vehicles.
5. Video surveillance.
6. Image alignment.

Conclusion: Translation Alignment estimates simple horizontal and vertical movement, whereas Parametric Motion represents complex movements using mathematical transformations. Both techniques are useful in dense motion estimation and video analysis.

Unit 3 • Q5

Evaluate Spline-Based Motion Models and Their Advantages.

> 📝 **Exam Question:** 5. Evaluate spline-based motion models and their advantages.

Unit III – Q5. Evaluate Spline-Based Motion Models and Their Advantages.

1. Definition of Spline-Based Motion Models

Spline-Based Motion Models are mathematical techniques used in Computer Vision to represent smooth and continuous object or camera movements.

They use piecewise polynomial functions called **Splines** to estimate motion between different image frames.

Unlike simple translation models, spline-based models can represent complex and non-linear movements.

2. Types of Spline-Based Motion Models

1. **Linear Spline:** Uses straight-line segments to represent movement between control points.
2. **Cubic Spline:** Uses third-degree polynomial functions to create smooth motion curves.
3. **B-Spline:** Uses control points and basis functions to represent flexible and smooth motion.
4. **Thin-Plate Spline (TPS):** Models smooth, non-rigid transformations and is useful for image deformation.

3. Working of Spline-Based Motion Models

The working process involves:

1. **Feature Detection:** Identify important points in consecutive image frames.
2. **Control Point Selection:** Select points that represent object movement.
3. **Spline Construction:** Apply spline functions to connect the control points smoothly.
4. **Motion Estimation:** Calculate intermediate positions using spline equations.
5. **Motion Interpolation:** Estimate smooth movement between frames.
6. **Final Output:** Generate a continuous motion representation.

4. Mathematical Representation

A cubic spline can be represented as:

[S(t)=a+bt+ct^2+dt^3]

Where:

- (S(t)) = Estimated position.
- (t) = Time parameter.
- (a,b,c,d) = Spline coefficients.

The coefficients are calculated using control points and smoothness conditions.

5. Advantages of Spline-Based Motion Models

1. **Smooth Motion:** Produces continuous and natural movement.
2. **Flexible Representation:** Handles complex and non-linear motion.
3. **Accurate Estimation:** Provides better motion representation between control points.

4. **Noise Reduction:** Smooth curves can reduce unwanted variations in estimated motion.
5. **Local Control:** Changing one control point generally affects only a limited part of the spline.
6. **Non-Rigid Motion:** Can represent movements such as facial expressions and body deformations.

6. Applications

1. Human motion tracking.
2. Video stabilization.
3. Facial expression recognition.
4. Animation and computer graphics.
5. Object tracking.
6. Medical image registration.

7. Limitations

- Requires proper selection of control points.
- Computational complexity can increase with more control points.
- Incorrect control points may produce inaccurate motion.

Conclusion: Spline-Based Motion Models provide smooth, flexible and continuous motion estimation. They are especially useful for complex and non-rigid movements in Computer Vision applications.

QUICK REVISION / COMPLETE RECAP

Unit 3: Motion Estimation & Structure from Motion (SfM) — Complete Recap

High-yield revision summary, definitions, key points to remember, and core formulas for last-minute exam prep.

1. Unit Overview & Summary

Unit 3 explores 3D geometric vision and motion estimation: estimating 3D camera trajectory and 3D scene structure from multiple photographs (Structure from Motion - SfM), dense motion estimation across frames, and parametric and spline-based non-rigid motion modeling.

2. Core Concepts & Definitions

- **Structure from Motion (SfM)**: The joint estimation of 3D scene point coordinates $\mathbf{X}_j$ and camera projection matrices $\mathbf{P}_i$ from a sequence of 2D overlapping images:

  FORMULA Governing Mathematical Equation

  $$\mathbf{x}_{ij} \sim \mathbf{P}_i \mathbf{X}_j = \mathbf{K}_i [\mathbf{R}_i \mid \mathbf{t}_i] \mathbf{X}_j$$

- **Epipolar Geometry**:

- **Baseline**: The 3D line connecting two camera optical centers $\mathbf{C}_1$ and $\mathbf{C}_2$.
- **Epipoles (e_1, e_2)**: Points where the baseline intersects each image plane.
- **Epipolar Plane**: Plane spanned by 3D point $\mathbf{X}$ and the two optical centers.
- **Epipolar Line**: Intersection of the epipolar plane with the image plane. Any point x in image 1 must lie on line $l' = Fx$ in image 2 (reduces 2D search to 1D search).

- **Essential Matrix (E)**: Operates on normalized calibrated image coordinates $(\hat{\mathbf{x}}' = \mathbf{K}'^{-1} \mathbf{x}')$:

  FORMULA Governing Mathematical Equation

  $$\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}, \quad \hat{\mathbf{x}}'^T \mathbf{E} \hat{\mathbf{x}} = 0$$

- Has 5 degrees of freedom (3 for rotation, 2 for direction of translation up to an unknown scale).

- Its two non-zero singular values are equal: $\text{diag}(\sigma, \sigma, 0)$.

- **Fundamental Matrix (F)**: Operates on raw, uncalibrated pixel coordinates:

  FORMULA Governing Mathematical Equation

  $$\mathbf{F} = \mathbf{K}'^{-T} \mathbf{E} \mathbf{K}^{-1}, \quad \mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$$

- Has 7 degrees of freedom (rank 2, $\det(\mathbf{F}) = 0$).

- Estimated using the **Normalized 8-Point Algorithm** (Hartley normalization: centroid at origin, mean distance $\sqrt{2}$).
- **Tomasi-Kanade Factorization Method**: Under affine / orthographic projection, visual tracking of n points across m frames forms a measurement matrix $\mathbf{W}$ of size $2m \times n$:

  FORMULA **Governing Mathematical Equation**

  $$\mathbf{W} = \mathbf{MS}$$

- $\mathbf{M}$ is the $2m \times 3$ camera motion matrix; $\mathbf{S}$ is the $3 \times n$ 3D structure matrix.
- By the Rank Theorem, $\text{rank}(\mathbf{W}) \le 3$.
- Factored directly using Singular Value Decomposition (SVD): $\mathbf{W} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$.
- **Triangulation**: Reconstructing a 3D point $\mathbf{X}$ from two known calibrated camera views $\mathbf{P}_1, \mathbf{P}_2$ and corresponding 2D points $\mathbf{x}_1, \mathbf{x}_2$ using Direct Linear Transformation (DLT) or midpoint ray intersection.
- **Bundle Adjustment**: The gold-standard non-linear optimization step in SfM that jointly refines camera parameters $\mathbf{P}_i$ and 3D point locations $\mathbf{X}_j$ by minimizing the total reprojection error:

  FORMULA **Governing Mathematical Equation**

  $$\min_{\{\mathbf{P}_i\},\{\mathbf{X}_j\}} \sum_{i=1}^{m} \sum_{j=1}^{n} v_{ij} \left\| \mathbf{x}_{ij} - \pi(\mathbf{P}_i, \mathbf{X}_j) \right\|^2$$

Solved using the **Levenberg-Marquardt (LM)** algorithm exploiting the sparse block structure (Schur complement). - **Dense Motion Estimation Hierarchy**: - *Translation* (2 DOF): $[x', y']^T = [x + u, y + v]^T$. - *Euclidean* (3 DOF): Rotation + Translation. - *Similarity* (4 DOF): Rotation + Translation + Scale. - *Affine* (6 DOF): Linear deformation + Translation. - *Planar Projective (Homography)* (8 DOF): Perspective projection of flat planar scenes. - **Spline-Based Motion Models**: Non-rigid deformation modeled by a grid of 2D B-spline control points (Free-Form Deformation - FFD), allowing smooth, locally controlled, non-rigid image warping and dense optical flow registration.

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3. Key Points to Remember for Exams

1. Epipolar geometry reduces the 2D correspondence search to a 1D search along the epipolar line.
2. In Two-Frame SfM, absolute scale cannot be recovered from visual images alone (requires GPS, IMU, or known distance baseline).
3. The Essential Matrix has rank 2 and 5 degrees of freedom; Fundamental Matrix has rank 2 and 7 degrees of freedom.
4. Triangulation ray intersection fails in the presence of noise; optimal triangulation minimizes reprojection distance.
5. Bundle Adjustment is called "Bundle" because it simultaneously optimizes the bundle of light rays from 3D points to camera centers.

✎ SHORT ANSWER QUESTIONS (2 MARKS)

Unit 3 — 2-Mark Exam Questions

Crisp, direct, point-to-point answers designed for maximum marks in university short-answer sections.

Q1 What is Structure from Motion (SfM)?

Structure from Motion (SfM) is a photogrammetric technique that reconstructs the 3D structure of a scene and estimates camera poses simultaneously from a series of overlapping 2D photographs captured from multiple viewpoints without prior knowledge of camera locations.

Q2 What is Bundle Adjustment?

Bundle Adjustment is an optimization technique used in 3D reconstruction and Structure from Motion (SfM). It simultaneously refines camera positions, camera orientations and 3D point coordinates. Its main objective is to minimize **reprojection error** between observed image points and projected 3D points. ## Computer Vision – 2 Marks Questions and Answers

Q3 What is Epipolar Geometry and the Essential Matrix?

Epipolar geometry is the intrinsic projective geometry between two camera views. The Essential Matrix (E = [t]_x * R) encapsulates the relative rotation R and translation t between two calibrated cameras, satisfying the epipolar constraint x'^T * E * x = 0 for corresponding normalized image points.

Q4 What is Dense Motion Estimation?

Dense motion estimation computes a continuous 2D displacement motion field (or optical flow) for every pixel between consecutive video frames using brightness constancy constraints, pixel alignment, and spatial regularization.

UNIT 4

Unit 4: Computational Photography, Image Stitching & Textures

Comprehensive Question & Answer Collection • Quick Revision Recap • 2-Mark Direct Questions

PART A: DETAILED EXAMINATION QUESTIONS & ANSWERS

Unit 4 • Q1

Describe Motion Analysis and Motion Models.

📝 **Exam Question:** **Unit-IV** 1\. Describe Motion Analysis and Motion Models.

FORMULA **Optical Flow Brightness Constancy Constraint Equation (BCCE)**

$$I_x u + I_y v + I_t = 0 \Leftrightarrow \nabla I^T \mathbf{v} + I_t = 0$$

Linearized first-order Taylor expansion relating spatial gradients (I_x, I_y) to optical velocity (u, v).

FORMULA **Lucas-Kanade Least-Squares Optical Flow Solution**

$$\begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix}$$

Solves the aperture problem over a local n x n neighborhood by inverting the 2x2 structure tensor.

Unit IV – Q1. Describe Motion Analysis and Motion Models.

1. Motion Analysis

Motion Analysis is the process of detecting, measuring and understanding the movement of objects or cameras across consecutive video frames.

It helps computers identify the position, direction, speed and movement patterns of objects over time.

Motion Analysis is widely used in video surveillance, robotics, autonomous vehicles and sports analysis.

2. Steps Involved in Motion Analysis

1. **Frame Acquisition:** Capture consecutive frames from a video.
2. **Preprocessing:** Remove noise and improve image quality.
3. **Motion Detection:** Identify changes between consecutive frames.
4. **Motion Estimation:** Calculate the direction and amount of movement.
5. **Object Tracking:** Follow moving objects across frames.
6. **Motion Interpretation:** Understand the movement and identify activities.

3. Motion Models

Motion Models are mathematical representations used to describe and estimate the movement of objects or cameras between consecutive frames.

The important types are:

1. Translation Model

Represents movement in horizontal and vertical directions without changing the object's shape or orientation.

Formula:

[x'=x+t_x,\quad y'=y+t_y]

Example: A car moving horizontally.

2. Rotation Model

Represents the turning of an object around a fixed point or centre.

Example: A rotating fan.

3. Affine Motion Model

Represents translation, rotation, scaling and shearing.

Formula:

[\begin{bmatrix}x'\y'\end{bmatrix} = \begin{bmatrix}a&b\c&d\end{bmatrix} \begin{bmatrix}x\y\end{bmatrix} + \begin{bmatrix}t_x\t_y\end{bmatrix}]

Example: Camera movement with rotation and scaling.

4. Projective Motion Model

Represents perspective changes in an image using a homography transformation.

Example: A road viewed from different camera angles.

5. Non-Rigid Motion Model

Describes movements in which an object's shape changes over time.

Example: Human walking, facial expressions and body movements.

4. Comparison of Motion Models

Model	Main Function
Translation	Changes position
Rotation	Changes orientation
Affine	Handles rotation, scaling and shearing
Projective	Handles perspective changes
Non-Rigid	Handles shape deformation

5. Applications of Motion Analysis

1. **Video Surveillance:** Detects suspicious movements.
2. **Autonomous Vehicles:** Tracks vehicles and pedestrians.
3. **Sports Analysis:** Studies player movements.
4. **Robotics:** Helps robots understand object movement.
5. **Medical Applications:** Analyzes human body movements.

Conclusion: Motion Analysis identifies and understands object movement, while Motion Models mathematically represent that movement. Both are important for object tracking, video processing and intelligent vision systems.

Unit 4 • Q2

Define and Explain Image Stitching Along with Global Alignment.

> 📝 **Exam Question:** 2. Define and explain image stitching along with global alignment.

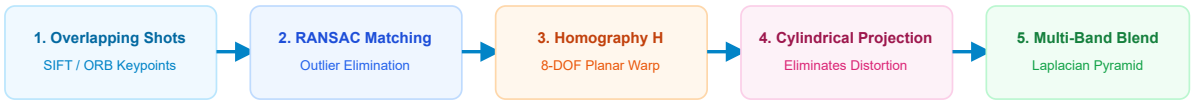


FIGURE 4.1: Panoramic Image Stitching Pipeline

Keypoints matched across photos estimate homography H via RANSAC, warps to cylindrical coordinates, and merges via multi-band blending.

Unit IV – Q2. Define and Explain Image Stitching Along with Global Alignment.

1. Definition of Image Stitching

Image Stitching is the process of combining two or more overlapping images to create a single large and continuous image.

It uses feature detection, feature matching, image alignment and blending techniques to produce a seamless panoramic image.

Example: Combining multiple photographs of a landscape to create a 360-degree panoramic view.

2. Steps Involved in Image Stitching

Step 1: Image Acquisition

Capture multiple images with overlapping regions.

Step 2: Feature Detection

Identify important keypoints using algorithms such as SIFT, SURF or ORB.

Step 3: Feature Matching

Find corresponding feature points between overlapping images.

Step 4: Homography Estimation

Calculate the transformation matrix to align the images.

[\begin{bmatrix}x'\y'\w'\end{bmatrix} = H\begin{bmatrix}x\y\1\end{bmatrix}]

Where (H) is a 3×3 Homography Matrix.

Step 5: Image Warping

Transform one image according to the estimated homography so that it aligns with another image.

Step 6: Blending

Combine overlapping regions smoothly to reduce visible boundaries.

Step 7: Final Panorama

Generate a single panoramic image.

3. Global Alignment

Global Alignment is the process of aligning all input images into a common coordinate system using estimated transformations.

Instead of aligning only two neighbouring images, it considers the relationships among all images to maintain overall consistency.

Working Steps:

1. Select a reference image.
2. Find feature correspondences between images.
3. Estimate transformation matrices.
4. Calculate the global positions of all images.
5. Optimize alignment to reduce errors.
6. Transform all images into a common coordinate system.

4. Importance of Global Alignment

1. **Consistency:** Maintains alignment across all images.
2. **Accuracy:** Reduces accumulated alignment errors.
3. **Seam Reduction:** Helps create smoother image boundaries.
4. **Better Panorama:** Produces a continuous panoramic image.
5. **Error Correction:** Helps minimize misalignment between images.

5. Difference Between Image Stitching and Global Alignment

Image Stitching	Global Alignment
Combines multiple images	Aligns all images together
Produces a panoramic image	Establishes common coordinates
Includes blending	Focuses on transformation consistency
Final image generation	Alignment optimization

6. Applications

- Panorama photography.
- Satellite image mapping.
- Medical image reconstruction.
- Virtual reality.
- Aerial photography.

Conclusion: Image Stitching combines overlapping images into one panoramic image, while Global Alignment ensures that all images are accurately positioned in a common coordinate system. Both are important for producing high-quality panoramic images.

Unit 4 • Q3

Explain Composing Surfaces and Blending Methods with Diagrams.

> 📝 **Exam Question:** 3. Explain about composing surfaces and blending methods with diagrams.

Unit IV – Q3. Explain Composing Surfaces and Blending Methods with Diagrams.

1. Composing Surfaces

Composing Surfaces is the process of combining multiple images or image regions onto a common surface to create a single, continuous visual representation.

In Computer Vision, it is commonly used in image stitching and panorama creation. Different images are transformed and placed on a common coordinate plane called the **Compositing Surface**.

Working Steps:

1. **Image Selection:** Select multiple overlapping images.
2. **Image Alignment:** Align images using feature matching and homography.
3. **Surface Creation:** Create a common canvas or compositing surface.
4. **Image Warping:** Transform images to fit the common surface.
5. **Image Placement:** Place the transformed images at their correct positions.
6. **Blending:** Combine overlapping regions smoothly.

Diagram:

Example: Combining three overlapping landscape photographs on a common canvas to create a panoramic image.

2. Blending Methods

Blending is the process of combining overlapping image regions to remove visible boundaries and create a smooth transition between images.

The important blending methods are:

1. Average Blending

- Calculates the average intensity of overlapping pixels.
- Simple and easy to implement.
- May produce ghosting if objects move between images.

Formula:

[I=\frac{I_1+I_2}{2}]

2. Feathering Blending

- Uses weighted averaging of overlapping image pixels.
- Gives more importance to pixels near the centre of each image.
- Reduces visible seams.

Formula:

[I=w_1I_1+w_2I_2]

Where (w_1+w_2=1).

3. Multi-Band Blending

- Separates images into different frequency bands.
- Blends low and high frequencies separately.
- Produces smooth transitions and preserves details.

4. Seam-Based Blending

- Finds an optimal boundary between overlapping images.
- Selects a suitable path to minimize visible differences.
- Useful when images contain different objects or lighting.

Diagram:

3. Comparison of Blending Methods

Method	Main Feature
Average	Simple pixel averaging
Feathering	Weighted averaging
Multi-Band	Frequency-based blending
Seam-Based	Finds optimal boundary

4. Applications

1. Panorama creation.
2. Satellite image stitching.
3. Medical image composition.
4. Virtual reality.
5. Aerial photography.

Conclusion: Composing Surfaces places multiple images on a common canvas, while Blending Methods combine overlapping regions smoothly. Both techniques are essential for creating seamless and high-quality panoramic images.

Unit 4 • Q4

Explain High Dynamic Range (HDR) Imaging in Computational Photography.

📝 **Exam Question:** 4. Explain High Dynamic Range (HDR) imaging in computational photography.

FORMULA **Debevec-Malik HDR Radiance Calibration Equation**

$$\ln g(Z_{ij}) = \ln E_i + \ln \Delta t_j$$

Relates pixel digital value Z_ij to scene irradiance E_i and exposure duration Δt_j via response curve g.

Unit IV – Q4. Explain High Dynamic Range (HDR) Imaging in Computational Photography.

1. Definition of HDR Imaging

High Dynamic Range (HDR) Imaging is a computational photography technique used to capture and combine multiple images taken at different exposure levels to produce a single image with greater details in both bright and dark areas.

It helps overcome the limitations of normal cameras when capturing scenes with large differences in brightness.

Example: Photographing a building where the sky is very bright but the building is dark.

2. Working of HDR Imaging

HDR imaging involves the following steps:

Step 1: Image Capture

Capture multiple photographs of the same scene using different exposure settings.

- Underexposed image: Captures details in bright areas.
- Normal exposure image: Captures balanced details.
- Overexposed image: Captures details in dark areas.

Step 2: Image Alignment

Align all captured images to ensure that objects appear in the same positions.

Step 3: Exposure Fusion

Combine useful information from all images to preserve details in bright and dark regions.

Step 4: HDR Image Creation

Generate a single image with a wider range of brightness information.

Step 5: Tone Mapping

Convert the HDR image into a display-friendly format while maintaining important details and contrast.

3. HDR Imaging Diagram

4. Mathematical Representation

HDR image formation can be represented as:

[HDR(x,y)=\sum_{i=1}^{n}w_i(x,y)I_i(x,y)]

Where:

- (I_i(x,y)) = Input image at different exposures.
- (w_i(x,y)) = Weight assigned to each image.
- (n) = Number of images.
- (HDR(x,y)) = Combined image information.

5. Importance and Advantages

1. **Better Details:** Preserves details in bright and dark areas.
2. **Improved Contrast:** Produces balanced brightness.
3. **Better Image Quality:** Creates visually rich photographs.
4. **Exposure Correction:** Reduces problems caused by incorrect exposure.
5. **Wide Dynamic Range:** Captures a wider range of light intensities.

6. Applications

- Landscape photography.
- Smartphone cameras.
- Medical imaging.
- Security cameras.
- Architectural photography.
- Computer vision systems.

7. Limitations

- Requires multiple images in traditional HDR capture.
- Moving objects may produce ghosting.
- Requires additional processing.

Conclusion: HDR Imaging combines images with different exposure levels to preserve details in bright and dark regions. It is an important computational photography technique for producing high-quality and realistic images.

Unit 4 • Q5

Explain How Super-Resolution Techniques Are Used for Blur Removal and Improving Image Quality.

> 📝 **Exam Question:** 5. Apply how the super-resolution techniques for blur removal and improving image quality.

Unit IV – Q5. Explain How Super-Resolution Techniques Are Used for Blur Removal and Improving Image Quality.

1. Definition of Super-Resolution

Super-Resolution (SR) is an image processing technique used to improve image resolution by converting low-resolution (LR) images into high-resolution (HR) images.

It increases image details, sharpness and clarity by recovering or estimating missing visual information.

It is useful for improving blurred, pixelated and low-quality images.

2. Working of Super-Resolution

The process involves the following steps:

Step 1: Input Image

Take a low-resolution or blurred image as input.

Step 2: Image Analysis

Analyze the image to identify missing details, edges and patterns.

Step 3: Feature Extraction

Extract important features such as edges, textures and shapes.

Step 4: Resolution Enhancement

Use interpolation or learning-based algorithms to estimate additional pixels and improve image resolution.

Step 5: Blur Reduction

Apply suitable restoration and sharpening techniques to improve image clarity.

Step 6: Output Image

Generate a high-resolution image with improved visual quality.

3. Types of Super-Resolution Techniques

1. Interpolation-Based Super-Resolution

- Estimates new pixel values using neighbouring pixels.
- Simple and fast.
- Examples: Bilinear and Bicubic Interpolation.

2. Reconstruction-Based Super-Resolution

- Combines information from multiple low-resolution images.
- Recovers missing image details.
- Useful in medical and satellite imaging.

3. Learning-Based Super-Resolution

- Uses Machine Learning and Deep Learning models.
- Learns image patterns from training data.
- Examples: SRCNN and EDSR.

4. Multi-Frame Super-Resolution

- Combines multiple images of the same scene.
- Uses additional information from different frames.
- Helps improve image details.

4. Mathematical Representation

The image degradation model is:

[LR=(HR*K)\downarrow_s+n]

Where:

- (LR) = Low-resolution image.
- (HR) = Original high-resolution image.
- (K) = Blur kernel.
- (\downarrow_s) = Downsampling operation.
- (n) = Noise.

Super-resolution attempts to estimate the original HR image from the degraded LR image.

5. Applications

1. Medical image enhancement.
2. Satellite image processing.
3. CCTV footage enhancement.
4. Face recognition.
5. Digital photography.
6. Video quality improvement.

6. Advantages

- Improves image resolution.
- Enhances visual details.
- Reduces pixelation.
- Can improve perceived sharpness.
- Helps in image analysis.

Conclusion: Super-Resolution improves low-resolution images by estimating missing details and increasing pixel resolution. Combined with deblurring techniques, it helps produce clearer and higher-quality images.

Unit 4 • Q6

Explain Image Matting and Compositing with a Neat Diagram.

Exam Question: 6. Explain with neat diagram about image Matting and Composting.



FIGURE 4.2: **Image Matting Model and Trimap Architecture**

Trimap guides optimization to solve the underconstrained 7-unknown matting equation.

FORMULA **Alpha Matting Compositing Formulation**

$$I = \alpha F + (1 - \alpha)B, \quad \alpha \in [0, 1]$$

Decomposes observed color I into foreground color F, background color B, and opacity matte α.

Unit IV – Q6. Explain Image Matting and Compositing with a Neat Diagram.

1. Image Matting

Image Matting is an image processing technique used to separate the foreground object from the background by estimating the transparency of each pixel.

It is especially useful for objects with complex boundaries, such as hair, glass, smoke and transparent objects.

Working Steps of Image Matting:

1. **Input Image:** Take an image containing foreground and background objects.
2. **Trimap Creation:** Divide the image into three regions:
3. Foreground: Known object pixels.
4. Background: Known background pixels.
5. Unknown: Pixels whose transparency must be estimated.
6. **Alpha Estimation:** Calculate the transparency value of unknown pixels.
7. **Foreground Extraction:** Separate the foreground object from the original background.
8. **Output:** Generate a foreground image with transparency information.

Matting Diagram:

Alpha Matting Formula:

[I=\alpha F+(1-\alpha)B]

Where:

- (I) = Original image.

- (F) = Foreground colour.
- (B) = Background colour.
- (\alpha) = Transparency value (0 to 1).

2. Image Compositing

Image Compositing is the process of combining two or more images or visual elements to create a single final image.

It uses the foreground extracted through matting and combines it with a new background.

Working Steps of Image Compositing:

1. **Foreground Selection:** Select the foreground object.
2. **Background Selection:** Choose a new background image.
3. **Image Alignment:** Adjust the size and position of the foreground.
4. **Blending:** Combine foreground and background using alpha values.
5. **Final Output:** Generate the combined image.

Compositing Diagram:

Compositing Formula:

[C=\alpha F+(1-\alpha)B]

Where:

- (C) = Final composite image.
- (F) = Foreground image.
- (B) = Background image.
- (\alpha) = Transparency value.

3. Difference Between Image Matting and Compositing

Image Matting	Image Compositing
Separates foreground from background	Combines multiple images
Estimates pixel transparency	Uses transparency for blending
Produces an alpha matte	Produces a final composite image
Used for object extraction	Used for background replacement

4. Applications

1. Photo editing and background replacement.
2. Film and video production.
3. Green-screen technology.
4. Augmented Reality.
5. Computer graphics.
6. Virtual backgrounds in video calls.

5. Advantages

- Separates complex foreground objects.
- Supports transparent objects.

- Creates realistic combined images.
- Improves visual editing quality.

Conclusion: Image Matting extracts foreground objects by estimating pixel transparency, whereas Image Compositing combines the extracted objects with new backgrounds. Both techniques are important in digital photography, video editing and computer graphics.

Unit 4 • Q7

Design a Method for Texture Analysis and Synthesis in Computer Vision Applications.

> 📝 **Exam Question:** 7. Design a method for texture and synthesis analysis in computer vision applications.

Unit IV – Q7. Design a Method for Texture Analysis and Synthesis in Computer Vision Applications.

1. Introduction

Texture Analysis and Synthesis are important techniques in Computer Vision used to understand, classify and generate image textures.

- **Texture Analysis:** Identifies and studies surface patterns, such as roughness, smoothness, repetition and regularity.
- **Texture Synthesis:** Generates new textures that look similar to an existing texture without simply copying it.

Examples: Identifying wood, fabric, grass, stone and brick textures.

2. Texture Analysis Method

Texture Analysis extracts useful information from an image to identify different surface patterns.

Working Steps:

1. **Image Acquisition:** Capture an image containing different textures.
2. **Preprocessing:** Remove noise and improve image quality.
3. **Feature Extraction:** Extract texture features such as contrast, energy, correlation and homogeneity.
4. **Feature Classification:** Classify textures using Machine Learning algorithms.
5. **Result Generation:** Identify and label different texture regions.

Common Techniques:

- **GLCM:** Measures relationships between neighbouring pixel intensities.
- **LBP:** Describes local texture patterns using binary comparisons.
- **Gabor Filter:** Detects texture patterns at different scales and orientations.

3. Texture Synthesis Method

Texture Synthesis is the process of generating a new image or larger texture using the patterns and features of a small sample texture.

Working Steps:

1. **Input Texture:** Select a small sample texture image.
2. **Pattern Analysis:** Analyze pixel relationships, colour and texture patterns.
3. **Patch Selection:** Select suitable image patches.
4. **Texture Generation:** Combine patches or generate new patterns.
5. **Seam Removal:** Reduce visible boundaries between patches.
6. **Final Output:** Produce a larger, realistic texture image.

Common Techniques:

1. **Pixel-Based Synthesis:** Generates pixels using neighbouring pixel information.

2. **Patch-Based Synthesis:** Combines small texture patches.
3. **Example-Based Synthesis:** Uses an existing texture as a reference.
4. **Deep Learning-Based Synthesis:** Uses neural networks to generate realistic textures.

4. Difference Between Texture Analysis and Synthesis

Texture Analysis	Texture Synthesis
Studies existing textures	Generates new textures
Extracts texture features	Reproduces texture patterns
Used for classification	Used for image generation
Example: Identifying wood	Example: Generating wood texture

5. Applications

1. **Medical Imaging:** Identifies tissue patterns.
2. **Agriculture:** Detects plant diseases.
3. **Remote Sensing:** Classifies land surfaces.
4. **Computer Graphics:** Generates realistic textures.
5. **Industrial Inspection:** Detects surface defects.
6. **Image Editing:** Creates seamless backgrounds.

6. Advantages

- Improves object and surface classification.
- Helps identify material properties.
- Generates realistic visual patterns.
- Supports automated image analysis.

Conclusion: Texture Analysis identifies and classifies image patterns, whereas Texture Synthesis generates new textures using existing patterns. Both techniques are useful in medical imaging, agriculture, computer graphics and industrial inspection.

QUICK REVISION / COMPLETE RECAP

Unit 4: Computational Photography, Image Stitching & Textures — Complete Recap

High-yield revision summary, definitions, key points to remember, and core formulas for last-minute exam prep.

1. Unit Overview & Summary

Unit 4 covers Computational Photography, advanced motion analysis, panoramic image stitching, seamless multi-band and gradient-domain compositing, High Dynamic Range (HDR) radiance capture, and texture modeling.

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2. Core Concepts & Definitions

- **Motion Analysis & Optical Flow**:
- **Brightness Constancy Constraint Equation (BCCE)**: Assumes a pixel's intensity does not change under small motion:

FORMULA Governing Mathematical Equation

$$I(x+u, y+v, t+1) = I(x, y, t) \Longrightarrow I_x u + I_y v + I_t = 0$$

- **Aperture Problem**: One equation with two unknowns (u, v). Only normal velocity (component parallel to gradient ∇I) can be recovered locally. - **Lucas-Kanade Method**: Solves aperture problem by assuming motion is constant within a small local $n \times n$ neighborhood:

FORMULA Governing Mathematical Equation

$$\begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum I_x I_t \\ \sum I_y I_t \end{bmatrix}$$

- **Horn-Schunck Method**: Global variational formulation combining BCCE with a spatial smoothness regularizer. - **Image Stitching (Panoramas)**: 1. *Feature Extraction*: Detect SIFT/ORB keypoints across overlapping photos. 2. *Feature Matching*: Match keypoints via nearest neighbor ratio test. 3. *Homography Estimation*: Compute 3×3 homography $\mathbf{H}$ using RANSAC to eliminate outliers. 4. *Cylindrical / Spherical Warping*: Projects planar photos onto cylinder/sphere to eliminate perspective distortion in wide-angle panoramas:

FORMULA Governing Mathematical Equation

$$x' = s \cdot \arctan(x / f), \quad y' = s \cdot \frac{y}{\sqrt{x^2 + f^2}}$$

5. *Global Alignment & Bundle Adjustment*: Eliminates cumulative rotational drift and closes the 360° loop. - **Compositing & Blending Methods**: - *Alpha Feathering*: Linear ramp weight transition along seam; prone to ghosting under parallax. - *Multi-Band Laplacian Pyramid Blending (Burt & Adelson)*: Decomposes images into

Laplacian pyramids. Low spatial frequencies blended across wide transition zones; high frequencies blended across narrow seams. - *Poisson Image Editing (Perez et al.)*: Operates in the gradient domain. Solves Poisson equation $\nabla^2 f = \text{div } \mathbf{v}$ with Dirichlet boundary conditions, guaranteeing seamless color and lighting integration without seam artifacts. - **High Dynamic Range (HDR) Imaging**: - Captures real-world scenes exceeding 100,000:1 dynamic range by combining multiple low dynamic range (LDR) exposures Δt_j. - **Debevec-Malik Algorithm**: Recovers non-linear camera response curve $g(Z)$ by solving:

FORMULA **Governing Mathematical Equation**

$$\min \sum_{i=1}^{N} \sum_{j=1}^{P} \left[w(Z_{ij}) (\ln g(Z_{ij}) - \ln E_i - \ln \Delta t_j) \right]^2 + \lambda \sum_{z=Z_{\min}+1}^{Z_{\max}-1} [w(z) g''(z)]^2$$

- **Tone Mapping**: Compresses 32-bit floating-point radiance map into displayable 8-bit dynamic range. - *Global Operators*: Apply identical non-linear curve (Reinhard tone mapping). - *Local Operators*: Compress base layer using bilateral filter while preserving fine local detail. - **Image Matting**: Extracts foreground element with partial transparency:

FORMULA **Governing Mathematical Equation**

$$I = \alpha F + (1 - \alpha) B, \quad \alpha \in [0, 1]$$

- Uses a **Trimap** dividing image into Known Foreground ($\alpha = 1$), Known Background ($\alpha = 0$), and Unknown Transition region ($0 < \alpha < 1$).
- **Texture Analysis & Synthesis**:
- *Statistical*: Gray-Level Co-occurrence Matrix (GLCM) computing Energy, Contrast, Homogeneity, and Correlation.
- *Spectral / Filter Banks*: Gabor filters tuned to specific spatial frequencies and orientations.
- *Synthesis*: Efros-Leung Markov Random Field (MRF) non-parametric patch/pixel sampling matching neighborhood appearances.

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3. Key Points to Remember for Exams

1. Optical flow assumes brightness constancy, small motion, and spatial coherence.
2. Homography matrix $\mathbf{H}$ has 8 degrees of freedom; requires a minimum of 4 non-collinear point correspondences (using DLT).
3. RANSAC prevents misaligned panoramas caused by moving objects or mismatched keypoints.
4. Laplacian pyramid blending prevents edge blurring while eliminating exposure mismatch seams.
5. In the matting equation, 3 equations (RGB) have 7 unknowns ($F_r, F_g, F_b, B_r, B_g, B_b, \alpha$), making matting an ill-posed underconstrained inverse problem.

✎ SHORT ANSWER QUESTIONS (2 MARKS)

Unit 4 — 2-Mark Exam Questions

Crisp, direct, point-to-point answers designed for maximum marks in university short-answer sections.

Q1 What is Image Stitching?

Image Stitching is the process of combining two or more overlapping images to create a single large image. It uses feature detection, image alignment and blending techniques. It is commonly used to create panoramic photographs and wide-angle views.

Q2 What is Image Matting?

Image Matting is the process of separating foreground objects from the background by estimating the transparency (Alpha) of each pixel. It helps extract objects with complex boundaries, such as hair and transparent objects. It is commonly used in photo editing and video effects.

Q3 What is Lumi-Graphs?

Lumi-Graphs is not a widely standardized computer vision term. If your syllabus refers to **Lumigraphs**, they are a technique for representing and rendering 3D scenes using multiple images captured from different viewpoints. They help generate realistic views from new camera positions.

Q4 What is High Dynamic Range (HDR) Imaging?

High Dynamic Range (HDR) imaging is a computational photography technique that merges multiple photographs taken at varying exposure times into a single image with a high dynamic range of luminance, capturing details in both dark shadows and bright highlights.

Q5 What is Multi-Band Image Blending?

Multi-band image blending (Laplacian pyramid blending) decomposes overlapping images into distinct spatial frequency bands, blends low frequencies smoothly across wide seams to eliminate photometric variations, and blends high frequencies sharply across narrow seams to preserve crisp textures.

UNIT 5

Unit 5: 3D Reconstruction, Rendering & Face Recognition

Comprehensive Question & Answer Collection • Quick Revision Recap • 2-Mark Direct Questions

PART A: DETAILED EXAMINATION QUESTIONS & ANSWERS

Unit 5 • Q1

Describe How 3D Reconstruction and Recognition Can Be Done Using Shape from X and Active Range Finding.

📝 **Exam Question:** unit v 1. Describe how 3D Reconstruction and Recognition can be done using Shape from X and Active Range Finding.

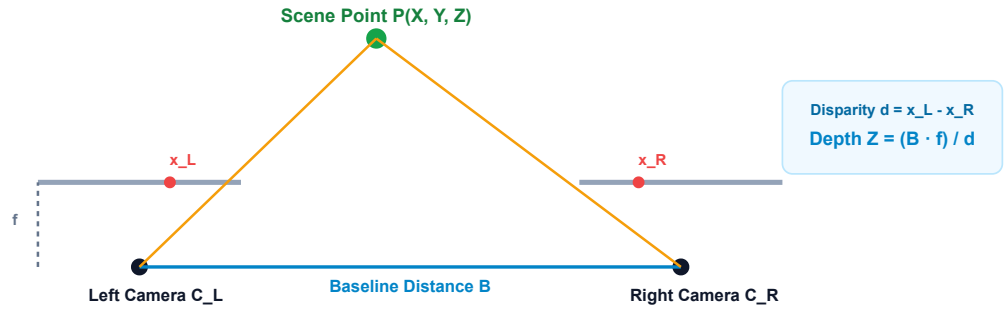


FIGURE 5.1: Stereo Vision Triangulation and Disparity Depth Recovery

Horizontal disparity d between corresponding points in calibrated parallel stereo cameras is inversely proportional to depth Z.

FORMULA **Stereo Disparity and Triangulated Depth Recovery**

$$d = x_L - x_R, \quad Z = \frac{B \cdot f}{d}$$

Horizontal disparity d along epipolar scanlines is inversely proportional to metric scene depth Z.

Unit V – Q1. Describe How 3D Reconstruction and Recognition Can Be Done Using Shape from X and Active Range Finding.

1. Introduction to 3D Reconstruction and Recognition

3D Reconstruction is the process of creating a three-dimensional model of an object or scene from 2D images or depth information.

3D Recognition is the process of identifying and classifying objects using their three-dimensional shape, size and geometric features.

Two important approaches are **Shape from X** and **Active Range Finding**.

2. Shape from X

Shape from X is a group of computer vision techniques used to estimate the 3D shape of an object from different visual clues available in 2D images.

Here, X represents different visual information.

Types of Shape from X:

1. **Shape from Shading:** Estimates surface shape using variations in brightness and lighting.
2. **Shape from Stereo:** Uses two images taken from different viewpoints to estimate depth.

3. **Shape from Motion:** Estimates 3D structure using multiple images captured while the camera or object moves.
4. **Shape from Texture:** Estimates surface orientation and depth using texture patterns.
5. **Shape from Silhouette:** Uses object outlines from different viewpoints to reconstruct its 3D shape.

Working Steps:

1. Capture images from different viewpoints.
2. Extract visual features such as edges, shading and texture.
3. Estimate depth and surface information.
4. Combine the estimated information.
5. Generate a 3D model.
6. Recognize the object by comparing its 3D features with stored models.

3. Active Range Finding

Active Range Finding is a technique that uses an external energy source, such as laser light or infrared light, to measure the distance between a sensor and an object.

It directly obtains depth information to create a 3D representation.

Common Methods:

1. **Time of Flight (ToF):** Measures the time taken by light to travel to an object and return.
2. **Structured Light:** Projects a known light pattern onto an object and analyzes its deformation.
3. **Laser Scanning:** Uses laser measurements to collect 3D surface points.

Working Steps:

1. Emit light or a known pattern towards the object.
2. Capture the reflected light using a sensor.
3. Calculate distance or depth information.
4. Generate a 3D point cloud.
5. Convert the point cloud into a 3D model.
6. Recognize the object using its geometric features.

4. Comparison

Shape from X	Active Range Finding
Uses visual information	Uses emitted energy
Often uses passive cameras	Uses active sensors
Estimates depth indirectly	Measures depth directly
Examples: Stereo, shading	Examples: ToF, laser scanning

5. Applications

1. Robotics and navigation.
2. Autonomous vehicles.
3. Medical imaging.
4. Industrial inspection.
5. Augmented Reality.

6. 3D object recognition.

Conclusion: Shape from X estimates 3D shape using visual clues, whereas Active Range Finding measures depth using emitted energy. Both methods are important for 3D reconstruction and object recognition in Computer Vision.

Unit 5 • Q2

Describe Point-Based Representation and Volumetric Representations in 3D Reconstruction.

> 📝 **Exam Question:** 2. Describe point-based representation and Volumetric Representations in 3D reconstruction.

Unit V – Q2. Describe Point-Based Representation and Volumetric Representations in 3D Reconstruction.

1. Introduction

3D Reconstruction is the process of creating a three-dimensional model of an object or scene using images, depth information or sensor data.

Two important methods used to represent 3D objects are **Point-Based Representation** and **Volumetric Representation**.

2. Point-Based Representation

Point-Based Representation is a 3D modelling technique in which an object is represented using a collection of individual points in three-dimensional space.

Each point contains coordinates (X, Y, Z) that describe its position.

Working Steps:

1. Capture images or depth information using cameras or sensors.
2. Extract 3D points from the captured data.
3. Assign coordinates (X, Y, Z) to each point.
4. Combine the points to form a point cloud.
5. Process and visualize the point cloud as a 3D representation.

Diagram:

Example: A 3D scanner captures thousands of points from a human face to create a 3D point cloud.

Advantages: - Simple representation. - Requires less memory than dense volumetric grids in many cases. - Suitable for large and complex scenes. - Easy to capture using 3D scanners.

Limitations: - Surface may contain gaps. - Points do not directly describe object connectivity. - Requires additional processing to create a continuous surface.

3. Volumetric Representation

Volumetric Representation is a 3D modelling technique in which an object is represented using small three-dimensional elements called **Voxels**.

A voxel is similar to a pixel but represents a volume instead of a 2D area.

Each voxel stores information such as occupancy, colour or density.

Working Steps:

1. Capture 3D information using cameras or sensors.
2. Divide the 3D space into a grid of voxels.
3. Assign values to each voxel.

4. Identify occupied and empty regions.
5. Combine occupied voxels to create a 3D model.

Diagram:

Example: Representing a human body using a 3D grid of voxels in medical imaging.

Advantages: - Represents complete 3D regions. - Handles complex object structures. - Useful for volume-based calculations. - Supports 3D medical visualization.

Limitations: - Requires high memory for large volumes. - Computationally expensive. - Fine details require higher voxel resolution.

4. Comparison

Point-Based	Volumetric
Uses individual 3D points	Uses 3D voxels
Represents surface points	Represents 3D space
Generally memory-efficient	Can require large memory
May contain surface gaps	Represents occupied volumes
Example: Point cloud	Example: Voxel grid

5. Applications

1. 3D object reconstruction.
2. Medical imaging.
3. Robotics and navigation.
4. Virtual Reality.
5. Autonomous vehicles.
6. Industrial inspection.

Conclusion: Point-Based Representation uses individual 3D coordinates to represent objects, whereas Volumetric Representation uses voxels to represent 3D space. Both methods are important for creating and analyzing 3D models in Computer Vision.

Unit 5 • Q3

Explain Image-Based Rendering Using Layered Depth Images (LDI).

> 📝 **Exam Question:** 3. Explain the image-based rendering using Layered Depth Images.

Unit V – Q3. Explain Image-Based Rendering Using Layered Depth Images (LDI).

1. Definition of Layered Depth Images (LDI)

Layered Depth Image (LDI) is an image-based rendering technique used in Computer Vision and Computer Graphics to represent a 3D scene using multiple depth layers for each pixel.

Unlike a normal image, which stores only the frontmost visible surface, LDI stores multiple colour and depth values along the same viewing direction.

It helps generate new views of a scene from different camera positions.

2. Structure of Layered Depth Images

An LDI contains the following information:

1. **Colour Information:** Stores the colour of each visible surface.
2. **Depth Information:** Stores the distance of each surface from the camera.
3. **Multiple Layers:** Stores information about different surfaces along the same viewing direction.
4. **Pixel Coordinates:** Represents the position of each pixel in the image.

3. Working of Image-Based Rendering Using LDI

The process involves the following steps:

Step 1: Image Acquisition

Capture images of a 3D scene using a camera or multiple viewpoints.

Step 2: Depth Estimation

Calculate the depth information of visible surfaces using depth sensors or image-based techniques.

Step 3: LDI Construction

Store multiple colour and depth samples for each pixel to create layered depth information.

Step 4: Viewpoint Transformation

Change the virtual camera position to generate a new viewpoint.

Step 5: Image Warping

Project the stored colour and depth information onto the new image plane.

Step 6: Hole Filling

Fill missing regions that become visible due to the camera movement using stored background layers or interpolation.

Step 7: Final Rendering

Generate the final image from the new camera viewpoint.

4. Advantages of LDI

1. **Multiple Surface Information:** Stores foreground and background details.
2. **Viewpoint Changes:** Generates images from different camera positions.
3. **Reduced Image Capture:** Can generate multiple views from stored image information.
4. **Realistic Rendering:** Helps preserve depth and scene appearance.
5. **Occlusion Handling:** Reveals previously hidden surfaces when their information is stored in deeper layers.

5. Applications

1. Virtual Reality (VR).
2. Augmented Reality (AR).
3. 3D scene visualization.
4. Interactive gaming.
5. Image-based rendering.
6. Virtual environment creation.

6. Limitations

- Requires accurate depth information.
- Large scenes may require significant storage.
- Missing depth information can create holes.
- Large viewpoint changes may reduce rendering quality.

Conclusion: Layered Depth Images store multiple colour and depth layers for each pixel. They allow new viewpoints to be generated through image warping and rendering, making them useful in VR, AR and 3D visualization.

Unit 5 • Q4

Explain Face Detection and Pedestrian Detection Techniques.

Exam Question: 5. Explain face detection and pedestrian detection techniques.

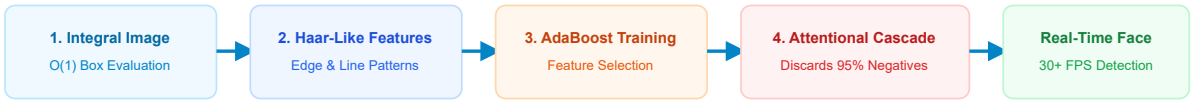


FIGURE 5.2: Viola-Jones Real-Time Face Detection Architecture

Combining integral images, AdaBoost weak classifier selection, and cascading stages yields real-time performance.

FORMULA **Viola-Jones Integral Image Constant-Time Box Sum**

$$ii(x,y) = \sum_{x \le x, y \le y} i(x,y), \quad extSum(D) = ii(4) + ii(1) - ii(2) - ii(3)$$

Allows rapid evaluation of rectangular Haar-like features in exactly 4 memory array lookups, O(1).

Unit V – Q5. Explain Face Detection and Pedestrian Detection Techniques.

1. Face Detection

Face Detection is a Computer Vision technique used to identify and locate human faces in images or videos. It detects facial regions regardless of their position, size or orientation.

It is widely used in security systems, smartphones and facial recognition applications.

Techniques of Face Detection:

1. **Haar Cascade Classifier:** Uses Haar-like features and a cascade of classifiers to detect faces quickly.
2. **HOG (Histogram of Oriented Gradients):** Extracts facial shape and edge information for detection.
3. **CNN (Convolutional Neural Network):** Uses deep learning to identify facial patterns and locate faces.
4. **MTCNN:** Detects faces and facial landmarks using a multi-stage neural network.

Working Steps:

1. Capture an image or video.
2. Preprocess the image.
3. Extract facial features.
4. Apply a face detection algorithm.
5. Locate faces using bounding boxes.
6. Display the detected faces.

2. Pedestrian Detection

Pedestrian Detection is a Computer Vision technique used to identify and locate people walking or standing in images and videos.

It helps intelligent systems recognize pedestrians and understand their movement.

Techniques of Pedestrian Detection:

1. **HOG + SVM:** Extracts gradient-based features using HOG and classifies pedestrians using Support Vector Machine.
2. **Background Subtraction:** Detects moving pedestrians by separating foreground objects from the background.
3. **CNN-Based Detection:** Uses deep learning models to detect pedestrians accurately.
4. **YOLO:** Detects pedestrians in real time by predicting bounding boxes and object classes.

Working Steps:

1. Capture images or video frames.
2. Preprocess the input.
3. Extract pedestrian features.
4. Apply a detection algorithm.
5. Identify pedestrians using bounding boxes.
6. Track pedestrians across frames if required.

3. Comparison Between Face and Pedestrian Detection

Face Detection	Pedestrian Detection
Detects human faces	Detects full human bodies
Focuses on facial features	Focuses on body shape and movement
Used in face-based systems	Used in traffic and surveillance
Example: Haar Cascade	Example: HOG + SVM
Detects facial regions	Detects human body regions

4. Applications

Face Detection: 1. Smartphone face unlock. 2. Security and surveillance. 3. Photo organization. 4. Attendance systems.

Pedestrian Detection: 1. Autonomous vehicles. 2. Traffic monitoring. 3. Video surveillance. 4. Pedestrian safety systems.

5. Advantages

- Automates human detection.
- Improves security and monitoring.
- Supports real-time visual analysis.
- Helps intelligent transportation systems.

Conclusion: Face Detection identifies human faces, whereas Pedestrian Detection identifies complete human bodies. Both are important Computer Vision techniques used in security, surveillance and autonomous systems.

Unit 5 • Q5

What are Eigenfaces? Explain Active Appearance and 3D Shape Models.

Exam Question: 6. What are Eigenfaces? Explain about Active appearance and 3D shape models.

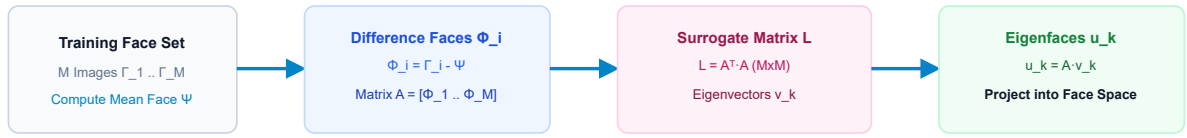


FIGURE 5.3: **Turk & Pentland Eigenfaces PCA Decomposition Pipeline**

Principal Component Analysis computes the principal dimensions of variation to project faces into a low-dimensional recognition space.

FORMULA **Turk & Pentland Surrogate Covariance Formulation for Eigenfaces**

$$\mathbf{C} = \frac{1}{M}\sum_{i=1}^{M} \Phi_i \Phi_i^T, \quad \mathbf{L} = \mathbf{A}^T\mathbf{A}, \quad \mathbf{u}_k = \mathbf{A}\mathbf{v}_k$$

Computes top eigenvectors of large N² x N² covariance matrix via tractable M x M matrix L.

Unit V – Q6. What are Eigenfaces? Explain Active Appearance and 3D Shape Models.

1. Eigenfaces

Eigenfaces is a face recognition technique based on **Principal Component Analysis (PCA)**. It represents human faces using important patterns and features extracted from a collection of face images.

Instead of comparing every pixel, Eigenfaces identifies the major variations among faces and represents each face using a smaller set of numerical values.

Working Steps:

1. Collect several face images of different people.
2. Convert all images into grayscale and resize them to the same dimensions.
3. Convert each image into a numerical vector.
4. Calculate the average face.
5. Apply PCA to find the important eigenvectors.
6. These eigenvectors are called **Eigenfaces**.
7. Represent a new face using weights corresponding to the Eigenfaces.
8. Compare these weights with stored face data to identify the person.

Applications: - Face recognition systems. - Security and authentication. - Human identification. - Digital image analysis.

2. Active Appearance Models (AAM)

Active Appearance Model (AAM) is a computer vision technique used to model and detect the shape and appearance of an object, especially a human face.

It combines two important models:

1. **Shape Model:** Represents the positions of facial landmarks such as eyes, nose, mouth and jawline.
2. **Appearance Model:** Represents the texture, colour and intensity information of the face.

Working Steps:

1. Collect training face images.
2. Mark important facial landmarks.
3. Build a statistical shape model.
4. Extract and model facial appearance.
5. Combine shape and appearance models.
6. Fit the model to a new image to locate facial features.

Applications: - Facial expression analysis. - Face tracking. - Facial feature detection. - Medical image analysis.

3. 3D Shape Models

3D Shape Models represent objects or human faces using three-dimensional information: width, height and depth.

Unlike 2D models, they describe the actual 3D structure of an object. They can represent facial features such as the nose, eyes, cheeks and jawline.

Working Steps:

1. Capture images from different viewpoints or use a 3D scanner.
2. Estimate the depth information.
3. Identify important 3D points.
4. Construct a 3D surface or mesh.
5. Apply texture and colour information.
6. Use the model for recognition or visualization.

Applications: - 3D face recognition. - Animation and gaming. - Virtual reality. - Medical and facial reconstruction.

4. Comparison

Model	Main Purpose	Important Feature
Eigenfaces	Face recognition	PCA and eigenvectors
Active Appearance Model	Face shape and texture modelling	Shape + appearance
3D Shape Model	Three-dimensional representation	Width, height and depth

Conclusion: Eigenfaces uses PCA for face recognition, Active Appearance Models combine facial shape and texture, and 3D Shape Models represent objects using three-dimensional information. These techniques are important in face recognition, tracking and computer vision applications.

Unit 5 • Q6

Design a Computer Vision System for Face Recognition Using Active Appearance Models.

📝 **Exam Question:** 7. Design a computer vision system for face recognition using active appearance models.



FIGURE 5.4: Active Appearance Model (AAM) Dual Synthesis Pipeline

AAM combines shape geometry deformation with photographic pixel appearance, iteratively minimizing synthesis error.

FORMULA **Active Appearance Model (AAM) Dual Shape & Texture Decomposition**

$$\mathbf{s} = \bar{\mathbf{s}} + \mathbf{P}_s\mathbf{p}, \quad \mathbf{g} = \bar{\mathbf{g}} + \mathbf{P}_g\mathbf{b}$$

Linear PCA parameterization of deformable landmark shape mesh s and shape-normalized texture g.

Unit V – Q7. Design a Computer Vision System for Face Recognition Using Active Appearance Models.

1. Introduction

An **Active Appearance Model (AAM)** is a Computer Vision technique that combines facial shape and appearance information to detect and analyse human faces.

A face recognition system using AAM identifies a person by comparing the facial features of an input image with previously stored face data.

2. System Architecture

The proposed system consists of the following major stages:

System Flow:

Image Acquisition → Face Detection → Preprocessing → Landmark Detection → AAM Construction → Feature Extraction → Face Matching → Recognition Result

3. Main Components and Working

Step 1: Image Acquisition

- Capture face images using a camera.
- Collect multiple face images of registered people.
- Store the images in a database.

Step 2: Face Detection

- Detect the face region from the input image.
- Remove unnecessary background information.
- Algorithms such as Haar Cascade or CNN can be used.

Step 3: Image Preprocessing

- Convert images into grayscale if required.
- Resize images to a standard size.
- Normalize brightness and contrast.
- Align faces to improve recognition accuracy.

Step 4: Facial Landmark Detection

Identify important facial points, such as:

- Eyes
- Nose
- Mouth
- Eyebrows
- Jawline

These points are used to construct the facial shape model.

Step 5: Active Appearance Model Construction

AAM combines two models:

1. **Shape Model:** Represents facial landmark positions and geometric structure.
2. **Appearance Model:** Represents facial texture, colour and intensity patterns.

The system combines both models to represent facial variations.

Step 6: Feature Extraction

The AAM extracts important facial shape and appearance information. It generates numerical parameters representing the input face.

Step 7: Face Matching and Recognition

- Compare the extracted features with stored face templates.
- Calculate the similarity between the input face and database faces.
- Select the closest matching identity if it satisfies a predefined threshold.
- Otherwise, classify it as unknown.

4. Algorithm

1. Start the system.
2. Capture an input face image.
3. Detect and preprocess the face.
4. Identify facial landmarks.
5. Fit the Active Appearance Model.
6. Extract shape and appearance parameters.
7. Compare parameters with stored templates.
8. Display the matching identity or unknown result.

9. Stop.

5. Advantages

- Combines facial shape and texture information.
- Models different facial expressions.
- Helps locate facial landmarks accurately.
- Useful for face tracking and recognition.

6. Limitations

- Requires suitable training images.
- Lighting and image quality can affect performance.
- Large facial variations may reduce accuracy.
- Model fitting can require significant computation.

7. Applications

1. Attendance management systems.
2. Security and access control.
3. Identity verification.
4. Human-computer interaction.
5. Face tracking and expression analysis.

Conclusion: A face recognition system using Active Appearance Models combines facial shape and appearance features to identify individuals. By detecting landmarks, constructing statistical models and matching extracted features with stored templates, the system can recognize faces efficiently.

If you want, I can:

- Describe system deployment challenges and mitigation strategies
- Explain training data preparation for AAM face recognition
- Discuss integration of AAM with deep learning for enhanced recognition

QUICK REVISION / COMPLETE RECAP

Unit 5: 3D Reconstruction, Rendering & Face Recognition — Complete Recap

High-yield revision summary, definitions, key points to remember, and core formulas for last-minute exam prep.

1. Unit Overview & Summary

Unit 5 covers high-level 3D Computer Vision, Volumetric and Point-Based Representations, Image-Based Rendering, Object Detection (Viola-Jones, HOG+SVM), Face Modeling (Eigenfaces, Active Appearance Models - AAM), and 3D Morphable Models (3DMM).

2. Core Concepts & Definitions

- **3D Reconstruction via Shape from X**:
- *Shape from Shading (SfS)*: Recovers surface orientation (surface normals $\mathbf{n}$) from photometric shading variations under known reflectance and lighting.
- *Shape from Stereo*: Reconstructs depth $Z = \frac{B \cdot f}{d}$ from horizontal disparity $d = x_L - x_R$ using calibrated stereo baseline B.
- *Shape from Silhouette / Visual Hull*: Reconstructs maximal 3D volume consistent with 2D foreground silhouettes viewed from multiple calibrated cameras.
- **Active Range Finding**:
- *Structured Light*: Projects known coded light patterns (binary stripes, phase-shifted sine fringes) and computes depth via triangulation distortion.
- *Time-of-Flight (ToF)*: Measures round-trip transit time Δt of emitted light pulses ($d = \frac{c \Delta t}{2}$).
- *LiDAR*: High-speed rotating laser range scanner producing dense 3D point clouds.
- **3D Geometry Representations**:
- *Point-Based*: Point Clouds (unorganized (X, Y, Z) coordinates) and Surfels (surface elements with position, normal vector, color, and local radius).
- *Volumetric*:
    - Voxel Grids: Regular 3D discrete lattice of volume elements.
    - Octrees: Hierarchical 8-ary tree dividing non-empty spatial volumes recursively, saving memory.
    - Truncated Signed Distance Functions (TSDF): Encodes distance from voxel to nearest surface; zero-crossing isosurface extracted via **Marching Cubes**.
- **Image-Based Rendering (IBR) & Layered Depth Images (LDI)**:
- Generates novel photographic views directly from sampled images without generating explicit complex polygon meshes.
- *Layered Depth Image (LDI)*: An image where each pixel contains a list of layered samples with depth, color, and normals, storing surfaces occluded from the primary viewpoint.
- Rendered via McMillan's 3D list-ordering warping algorithm, preventing self-occlusion artifacts, followed by splatting for hole-filling.
- **Viola-Jones Face Detection Algorithm**:
- *Integral Image*: Allows sum of pixel intensities within any rectangular region to be computed in constant time $O(1)$ using only 4 array lookups.
- *Haar-like Features*: Edge, line, and four-rectangle features capturing intensity differences between facial regions (eyes darker than cheeks, bridge of nose brighter than sides).
- *AdaBoost Feature Selection*: Selects critical features from over 180,000 candidates and constructs a strong classifier as a linear combination of weak classifiers.

- *Attentional Cascade*: Eliminates 95% of non-face background windows in the first few stages, achieving real-time 30+ FPS detection.
- **Pedestrian Detection (Dalal & Triggs HOG + Linear SVM)**:
- Computes dense Histogram of Oriented Gradients (HOG) over 8×8 pixel cells.
- Normalizes histograms across overlapping 16×16 blocks to ensure robust invariance to illumination and clothing shadows.
- Classifies feature vector using a linear Support Vector Machine (SVM).
- **Eigenfaces (Turk & Pentland Face Recognition)**:
- Applies Principal Component Analysis (PCA) to center-subtracted face images $\Phi_i = \mathbf{\Gamma}_i - \mathbf{\Psi}$.
- The covariance matrix $\mathbf{C} = \frac{1}{M} \sum_{i=1}^M \Phi_i \Phi_i^T$ of size $N^2 \times N^2$ is computationally intractable ($N^2 = 65,536$).
- Solved by finding eigenvectors $\mathbf{v}_k$ of surrogate $M \times M$ matrix $\mathbf{L} = \mathbf{A}^T \mathbf{A}$ ($M \ll N^2$), then calculating eigenfaces: $\mathbf{u}_k = \mathbf{A} \mathbf{v}_k$.
- Faces projected into low-dimensional "Face Space"; recognition performed via Euclidean distance to known face classes.
- **Active Appearance Models (AAM - Cootes et al.)**:
- Combined statistical generative model parameterizing shape deformations $\mathbf{s} = \bar{\mathbf{s}} + \mathbf{P}_s \mathbf{p}$ and appearance/texture variations $\mathbf{g} = \bar{\mathbf{g}} + \mathbf{P}_g \mathbf{b}$.
- Warps texture to shape-normalized reference mesh and iteratively optimizes shape and texture parameters to fit unseen target faces.

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3. Key Points to Remember for Exams

1. Disparity is inversely proportional to depth: objects at infinity have zero disparity; close objects have large disparity.
2. Viola-Jones integral image computes box sum as: $D - B - C + A$.
3. Viola-Jones cascade optimizes for speed: early stages have high detection rate (99.9%) and moderate false positive rate (50%), discarding negative windows instantly.
4. Eigenfaces represent global facial variation; sensitive to pose, scale, and extreme expression variations.
5. AAM combines both shape geometry and photographic texture, making it robust for landmark tracking and expression analysis.

✎ SHORT ANSWER QUESTIONS (2 MARKS)

Unit 5 — 2-Mark Exam Questions

Crisp, direct, point-to-point answers designed for maximum marks in university short-answer sections.

Q1 What is 3D Reconstruction?

3D Reconstruction is the process of creating a three-dimensional model of an object or scene from multiple 2D images. It uses techniques such as feature matching, triangulation and camera calibration. It is used in robotics, medical imaging, gaming and virtual reality.

Q2 What is 3D Object Recognition?

3D Object Recognition is the process of identifying and classifying objects using their three-dimensional shape, size and geometric features. It helps computers understand objects from different viewpoints. Applications include robotics, autonomous vehicles, augmented reality and industrial inspection.

Q3 What is Augmented Reality?

Augmented Reality (AR) is a technology that combines computer-generated objects, images or information with the real-world environment. It enhances the user's real-world experience using cameras, sensors and computer vision. Examples include Snapchat filters, Pokémon GO and AR navigation.

Q4 What are Eigenfaces?

Eigenfaces are the principal eigenvectors derived from Principal Component Analysis (PCA) performed on a large covariance matrix of facial images. They represent the primary dimensions of facial variance (lighting, features, expressions) and allow compact face representation and recognition.

Q5 What is an Active Appearance Model (AAM)?

An Active Appearance Model (AAM) is a statistical computer vision algorithm that jointly models shape deformations and textural appearances of deformable objects (such as human faces) using linear PCA models, iteratively matching model parameters to unseen target images.

🏆 OVERALL SUBJECT SUMMARY

Final Comprehensive Quick Revision Index (All Units)

Master topic matrix, mathematical formulation cross-reference, and canonical algorithm index across all 5 units.

Master Comprehensive Examination Cheat-Sheet & Topic Matrix

Unit	Title	Core Mathematical Formulation	Canonical Algorithms & Methods	High-Yield Exam Topics
I	**Image Formation & Enhancement**	Perspective Projection: $x = f\frac{X}{Z}$, $y = f\frac{Y}{Z}$ 2D DFT: $F(u,v) = \sum \sum f(x,y) e^{-j2\pi(ux/M + vy/N)}$ Wiener Filter: $W(u,v) = \frac{H^*}{\|H\|^2 + K}$	Pinhole camera model, Homogeneous transformations, Histogram Equalization, Linear/Non-linear spatial filtering, Markov Random Fields (MRF).	Geometric Transformations (3/4/6/8 DOF), 2D DFT significance, Histogram equalization steps, Inverse vs Wiener filtering, Spatial filter types.
II	**Segmentation & Feature Extraction**	N-Cut: $\text{Ncut}(A,B) = \frac{\text{cut}(A,B)}{\text{assoc}(A,V)} + \frac{\text{cut}(A,B)}{\text{assoc}(B,V)}$ Harris: $R = \det(M) - k(\text{trace}(M))^2$ SIFT: 128-D vector (4×4 grid $\times$ 8 bins)	Boykov-Kolmogorov Graph Cut, Normalized Cuts spectral solver, Mean Shift mode seeking, Canny Edge Detector (5 steps), SIFT, SURF, KLT feature tracking.	Graph Cut vs Normalized Cut, 5 steps of Canny edge detection, Feature matching via NNDR ratio test, SIFT vs SURF vs Harris comparison, Split and Merge.
III	**Motion & Structure from Motion**	Essential Matrix: $\mathbf{E} = [\mathbf{t}]\times \mathbf{R}$ *(5 DOF)* *Fundamental Matrix: $\mathbf{x}'^T \mathbf{F} \mathbf{x} = 0$ (7 DOF)* *Bundle Adjustment: $\min \sum \|\mathbf{x}_{ij} - \pi(\mathbf{P}_i, \mathbf{X}_j)\|^2$*	Normalized 8-Point Algorithm, Tomasi-Kanade SVD Factorization ($W = MS$, rank 3), DLT Triangulation, Levenberg-Marquardt Bundle Adjustment, Dense optical flow alignment, B-spline FFD warping.	Epipolar geometry elements (baseline, epipoles, epipolar lines), Two-Frame vs Factorization SfM, Triangulation & Bundle adjustment role, Motion estimation models.
IV	**Computational Photography**	Optical Flow BCCE: $I_x u + I_y v + I_t = 0$ Matting: $I = \alpha F + (1-\alpha)B$ Debevec HDR: $\ln g(Z_{ij}) = \ln E_i + \ln \Delta t_j$	Lucas-Kanade local optical flow, Homography RANSAC, Cylindrical panorama warping, Laplacian pyramid multi-band blending, Poisson gradient editing, Closed-form matting, Efros-Leung texture synthesis.	Aperture problem in optical flow, Image stitching pipeline, Feathering vs Multi-band vs Poisson blending, HDR recovery & Tone mapping, Trimap alpha matting.

Unit	Title	Core Mathematical Formulation	Canonical Algorithms & Methods	High-Yield Exam Topics
V	**3D Reconstruction & Face Recognition**	Stereo Depth: $Z = \frac{B \cdot f}{d}$ Eigenfaces: $\mathbf{C} = \frac{1}{M} \sum \Phi_i \Phi_i^T$, $\mathbf{L} = \mathbf{A}^T\mathbf{A}$ AAM: $\mathbf{s} = \bar{\mathbf{s}} + \mathbf{P}_s\mathbf{p}$, $\mathbf{g} = \bar{\mathbf{g}} + \mathbf{P}_g\mathbf{b}$	Shape from X (Stereo, Shading, Silhouette), LiDAR & ToF, TSDF & Marching Cubes, Layered Depth Images (LDI), Viola-Jones (Haar, AdaBoost, Cascade), HOG+SVM pedestrian detection, Turk-Pentland Eigenfaces, Active Appearance Models (AAM).	Shape from X methods, Point-based vs Volumetric 3D representations, Layered Depth Images, Viola-Jones 4 core stages, Eigenfaces PCA derivation, End-to-end AAM Face Recognition System design.